

Supplementary Material for Variance-Corrected Multi-Asset Equity Simulation with Hybrid Hidden Markov Marginals

S1. Empirical cross-term diagnostic

The variance correction of Eq. (8) assumes $\text{Cov}(\tilde{g}_i^c, g_m) \approx 0$, allowing the variance derivation to omit the cross-term. This assumption does not follow automatically from the fitting procedure because each JumpHMM uses the asset’s full growth-rate history, including its market-correlated component. A remaining market loading in the generator draw would invalidate the closed-form scale. We tested the assumption by sampling $R = 100$ generator growth-rate paths $\tilde{g}_i^{(r)}$ per ticker from each generator and computing the normalized cross-covariance against the real return path for a broad-market exchange-traded fund (ticker SPY):

$$\text{cov}_i^{\text{norm}} = \frac{1}{R} \sum_{r=1}^R \frac{\text{Cov}((\tilde{g}_i^{(r)})^c, g_m)}{\sigma_{\text{gen},i} \sigma_m}. \quad (\text{S1})$$

Centering does not change covariance, so these values are numerically identical to those obtained from the raw draws. We plotted the per-ticker values as histograms (Supplementary Fig. S1). Across the 423 tickers, the full-return JumpHMM generator (used by the hybrid and naive composition methods), the residual-fit JumpHMM generator (used by the JumpHMM-on-residuals composition method), and the Gaussian baseline had universe-mean normalized cross-covariances of 1.53×10^{-4} , 6.58×10^{-5} , and 2.26×10^{-4} , respectively. Their per-ticker values ranged from -0.0061 to 0.0061 . These values quantify the cross-term omitted in Eq. (8).

S2. Rebalancing under independent residuals

To determine what a portfolio-level use of these paths inherits from the independent-residual construction, this appendix derives the diversification return of a rebalanced portfolio under Eq. (4). The variance correction in Eq. (8) sets each synthetic asset’s second moment to its generator target. When that target matches the empirical second moment, the second-order formula for the diversification return is matched at the asset level as well. What the construction does not match is the cross-asset residual covariance, and the derivation below isolates where that omission enters.

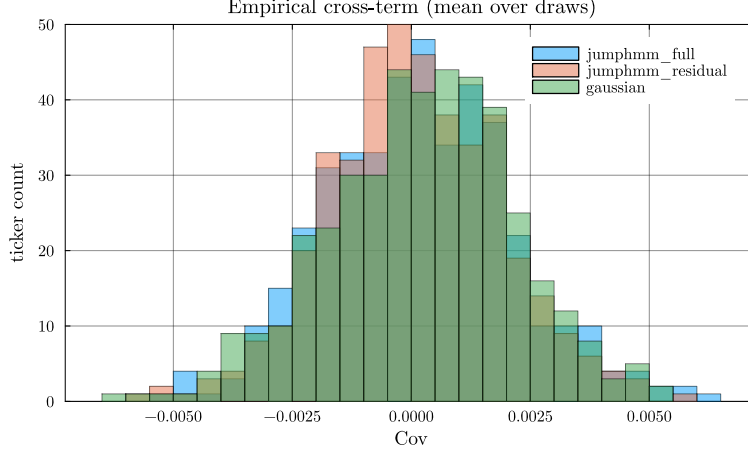


Figure S1: **Empirical normalized cross-covariance** $\text{Cov}(\tilde{g}_i^c, g_m)/(\sigma_{\text{gen},i} \sigma_m)$ **across the 423-ticker universe.** Histograms were stacked per generator, with each per-ticker value averaged over $R = 100$ generator growth-rate paths. Universe means were 1.53×10^{-4} , 6.58×10^{-5} , and 2.26×10^{-4} for the full-return JumpHMM, residual-fit JumpHMM, and Gaussian generators.

The synthetic paths reproduce per-asset marginals and single-factor cross-sectional dependence. Each residual generator retains its own temporal structure, but the residual paths are generated independently across assets at each date. The construction therefore omits cross-asset residual covariance. Consider a daily-rebalanced portfolio of N assets with constant target weights $w_i \geq 0$, $\sum_i w_i = 1$, reset to those weights at the start of each trading day. Throughout this derivation $g_i(t)$ is the annualized log-growth rate of Eq. (1), so the per-period log return on asset i is $g_i(t) \Delta t$ with $\Delta t = 1/252$ yr. The wealth process is:

$$W_t = W_{t-1} \sum_i w_i \exp(g_i(t) \Delta t), \quad (\text{S2})$$

so the per-step log return on the portfolio is $\log(W_t/W_{t-1}) = \log \sum_i w_i \exp(g_i(t) \Delta t)$. We suppress the time argument, writing g_i for $g_i(t)$. A second-order Taylor expansion of the exponential about zero is $\exp(g_i \Delta t) = 1 + g_i \Delta t + \frac{1}{2}(g_i \Delta t)^2 + O((g \Delta t)^3)$. Substituting it into each summand of $\sum_i w_i \exp(g_i \Delta t)$ gives:

$$\begin{aligned} \sum_i w_i \exp(g_i \Delta t) &= \sum_i w_i [1 + g_i \Delta t + \frac{1}{2}(g_i \Delta t)^2 + O((g \Delta t)^3)] & (\text{Taylor}) \\ &= \sum_i w_i + \Delta t \sum_i w_i g_i + \frac{1}{2}(\Delta t)^2 \sum_i w_i g_i^2 + O((g \Delta t)^3) & (\text{distribute}) \\ &= 1 + \Delta t \sum_i w_i g_i + \frac{1}{2}(\Delta t)^2 \sum_i w_i g_i^2 + O((g \Delta t)^3). & (\sum_i w_i = 1) \end{aligned} \quad (\text{S3})$$

A second-order Taylor expansion of $\log(1+x)$ about $x=0$ gives $\log(1+x) = x - \frac{1}{2}x^2 + O(x^3)$. Identifying $1+x$ with the right-hand side of Eq. (S3), the

order- Δt correction is $x := \Delta t \sum_i w_i g_i + \frac{1}{2}(\Delta t)^2 \sum_i w_i g_i^2$, and squaring keeps only the leading $(\Delta t)^2$ term: $x^2 = (\Delta t)^2 (\sum_i w_i g_i)^2 + O((\Delta t)^3)$ because the cross term has an extra Δt factor. Substitution gives:

$$\begin{aligned}
\log \frac{W_t}{W_{t-1}} &= \log \left(1 + x + O((g \Delta t)^3) \right) \\
&= x - \frac{1}{2} x^2 + O(x^3) \\
&= \Delta t \sum_i w_i g_i + \frac{1}{2} (\Delta t)^2 \sum_i w_i g_i^2 \\
&\quad - \frac{1}{2} (\Delta t)^2 \left(\sum_i w_i g_i \right)^2 + O((g \Delta t)^3) \\
&= \Delta t \sum_i w_i g_i + \frac{1}{2} (\Delta t)^2 \left[\sum_i w_i g_i^2 - \left(\sum_i w_i g_i \right)^2 \right] + O((g \Delta t)^3). \tag{S4}
\end{aligned}$$

Dividing through by Δt extracts the portfolio's annualized one-step log-growth rate $g_p(t) := \Delta t^{-1} \log(W_t/W_{t-1})$, the portfolio counterpart of the per-asset $g_i(t)$ (still suppressing the time argument):

$$g_p = \sum_i w_i g_i + \frac{1}{2} \Delta t \left(\sum_i w_i g_i^2 - \left(\sum_i w_i g_i \right)^2 \right) + O((\Delta t)^2). \tag{S5}$$

To obtain expected annualized portfolio growth, we restored the time argument and took unconditional expectations of Eq. (S5) under stationarity. Define the per-asset annualized mean and variance:

$$\mu_i := \mathbb{E}(g_i(t)), \quad \sigma_i^2 := \text{Var}(g_i(t)), \tag{S6}$$

the per-rate covariance $\Sigma_{ij} := \text{Cov}(g_i(t), g_j(t))$, and the per-rate portfolio variance $\sigma_p^2 := \text{Var}(\sum_i w_i g_i(t)) = w^\top \Sigma w$. The $O(\Delta t)$ correction in Eq. (S5) contains two squared quantities, $\sum_i w_i g_i(t)^2$ and $(\sum_i w_i g_i(t))^2$. Both expand by the second-moment identity, which holds for any random variable X with finite second moment:

$$\mathbb{E}(X^2) = \mathbb{E}(X)^2 + \text{Var}(X). \tag{S7}$$

Applying this identity to $X = g_i(t)$ and summing against w_i gives:

$$\begin{aligned}
\mathbb{E} \left(\sum_i w_i g_i(t)^2 \right) &= \sum_i w_i \mathbb{E}(g_i(t)^2) && \text{(linearity)} \\
&= \sum_i w_i [\mu_i^2 + \sigma_i^2] && \text{(by Eq. (S7))} \\
&= \sum_i w_i \mu_i^2 + \sum_i w_i \sigma_i^2. \tag{S8}
\end{aligned}$$

Applying the identity to $X = \sum_i w_i g_i(t)$, whose mean is $\sum_i w_i \mu_i$ and whose variance is σ_p^2 , gives:

$$\begin{aligned}\mathbb{E}\left(\left(\sum_i w_i g_i(t)\right)^2\right) &= \mathbb{E}\left(\sum_i w_i g_i(t)\right)^2 + \text{Var}\left(\sum_i w_i g_i(t)\right) \quad (\text{by Eq. (S7)}) \\ &= \left(\sum_i w_i \mu_i\right)^2 + \sigma_p^2.\end{aligned}\tag{S9}$$

We substituted Eqs. (S8)–(S9) into the expectation of Eq. (S5) and grouped the variance and mean contributions as follows:

$$\begin{aligned}\mathbb{E}(g_p) &= \sum_i w_i \mu_i + \underbrace{\frac{1}{2} \Delta t \left(\sum_i w_i \sigma_i^2 - \sigma_p^2 \right)}_D \\ &\quad + \frac{1}{2} \Delta t \left[\sum_i w_i \mu_i^2 - \left(\sum_i w_i \mu_i \right)^2 \right] + O((\Delta t)^2),\end{aligned}\tag{S10}$$

the Booth-Fama identity for the daily-rebalanced portfolio in annualized growth-rate units (Booth and Fama, 1992; Markowitz, 1976). The middle term, D , is the diversification return: the $O(\Delta t)$ correction equal to half the difference between the weighted-average per-asset variance $\sum_i w_i \sigma_i^2$ and the portfolio variance σ_p^2 . The per-rate moments μ_i and σ_i^2 have units yr^{-1} and yr^{-2} respectively; the products $\Delta t \sigma_i^2$ and $\Delta t \sigma_p^2$ in Eq. (S10) are the annualized variances (yr^{-1}). The third term in Eq. (S10) is the cross-sectional dispersion of the per-asset annualized means, of order $\Delta t \mu^2$, against the variance term D of order $\Delta t \sigma^2$; because μ^2 lies more than two orders of magnitude below σ^2 for daily growth rates (Supplementary Table S1), we drop it. The realized annualized compound growth rate $\bar{g}_p = (T \Delta t)^{-1} \log(W_T/W_0) = T^{-1} \sum_t g_p(t)$ converges to $\mathbb{E}(g_p)$ by the ergodic theorem under any stationary residual process, so the long-run limit is fixed by the one-time-slice unconditional joint distribution of $(g_i(t))_i$ and does not, in itself, depend on the serial structure of the residual draw.

To rewrite D in a form that makes its non-negativity manifest, expand $\sigma_p^2 = \sum_i w_i^2 \sigma_i^2 + 2 \sum_{i < j} w_i w_j \Sigma_{ij}$ and use the identity $\sum_i w_i (1 - w_i) \sigma_i^2 = \sum_{i < j} w_i w_j (\sigma_i^2 + \sigma_j^2)$ that holds for any weights summing to one:

$$D = \frac{1}{2} \Delta t \left(\sum_i w_i \sigma_i^2 - \sigma_p^2 \right) = \frac{1}{2} \Delta t \sum_{i < j} w_i w_j \text{Var}(g_i - g_j) \geq 0, \tag{S11}$$

so D equals one-half the annualized weighted-average pairwise growth-rate-difference variance and is therefore non-negative, with equality only when every pair (i, j) comoves perfectly ($\text{Var}(g_i - g_j) = 0$) or all weight concentrates on a single asset. The rebalancing rule itself produces this effect: weights drift between rebalances, the rebalancer sells whatever has outperformed (its weight rose above target) and buys whatever has underperformed (its weight fell below target), and the trade systematically converts period-by-period cross-sectional

dispersion into compounded growth. Markowitz (1976) described this as the variance bonus of the long-run mean-variance allocation; Booth and Fama (1992) called it the diversification return.

To separate market and residual contributions to the diversification return, we substituted the single-index decomposition $g_i(t) = \alpha_i + \beta_i g_m(t) + \varepsilon_i(t)$, with market variance $\sigma_m^2 := \text{Var}(g_m(t))$, per-asset residual variance $\sigma_{\varepsilon,i}^2 := \text{Var}(\varepsilon_i(t))$, and $\text{Cov}(g_m, \varepsilon_i) = 0$ (verified in Sec. S1), into Eq. (S11) gives:

$$\text{Var}(g_i - g_j) = (\beta_i - \beta_j)^2 \sigma_m^2 + \text{Var}(\varepsilon_i - \varepsilon_j), \quad (\text{S12})$$

Because g_m and the residuals are uncorrelated, this relation splits the diversification return into market-beta-dispersion and idiosyncratic components:

$$D = \underbrace{\frac{1}{2} \Delta t \sigma_m^2 \sum_{i < j} w_i w_j (\beta_i - \beta_j)^2}_{D_{\text{mkt}}} + \underbrace{\frac{1}{2} \Delta t \sum_{i < j} w_i w_j \text{Var}(\varepsilon_i - \varepsilon_j)}_{D_{\varepsilon}}, \quad (\text{S13})$$

where the market-beta-dispersion component D_{mkt} equals $\frac{1}{2} \Delta t \sigma_m^2 [\sum_i w_i \beta_i^2 - (\sum_i w_i \beta_i)^2]$, the weighted cross-sectional β dispersion scaled by the annualized market variance $\Delta t \sigma_m^2$. Across the 423-ticker universe the calibrated β distribution spans $[0.03, 1.79]$ (Supplementary Fig. S4). The weighted variance of β is bounded by one quarter of its squared range, so $D_{\text{mkt}} \leq \frac{1}{8} \Delta t \sigma_m^2 (1.79 - 0.03)^2$; at the SPY annualized variance $\Delta t \sigma_m^2 = 0.030 \text{ yr}^{-1}$ of the closing-price series used for the multi-asset universe (Fig. 4), D_{mkt} is at most 0.012 yr^{-1} , or 1.2 percentage points per year over this universe. For the idiosyncratic component D_{ε} , residuals were approximately orthogonal across assets at the same time slice. Therefore, $\text{Var}(\varepsilon_i - \varepsilon_j) \approx \sigma_{\varepsilon,i}^2 + \sigma_{\varepsilon,j}^2$, and D_{ε} scales with the weighted-average annualized idiosyncratic variance $\Delta t \sigma_{\varepsilon,i}^2$ per name.

The second-order value of D depends on same-time unconditional moments, not on serial dependence. Equation (8) sets $\sigma_{\text{gen},i}^2 = \beta_i^2 \sigma_m^2 + \sigma_{\varepsilon,i}^2$ to its empirical target. Any residual generator with the same unconditional second moments therefore gives the same D . This includes a stationary first-order autoregressive [AR(1)] residual generator $\varepsilon_i(t) = \phi \varepsilon_i(t-1) + \eta_i(t)$ with independent and identically distributed innovations η_i tuned so that $\text{Var}(\varepsilon_i) = \sigma_{\eta,i}^2 / (1 - \phi^2) = \sigma_{\varepsilon,i}^2$. The parameter ϕ leaves the marginal distribution at a single time slice unchanged. It therefore does not change $\mathbb{E}(g_p)$ in Eqs. (S10) and (S13), but it changes sampling variance. For a stationary AR(1) residual with variance σ_{ε}^2 , the variance of its sample mean is approximately $T^{-1} \sigma_{\varepsilon}^2 (1 + \phi) / (1 - \phi)$. Under ergodicity, serial dependence changes per-path sampling noise, not the stationary mean. Serial dependence in the residual draw is therefore not a lever on the diversification return at the level of the second-order formula.

The direct structural difference is contemporaneous. The derivation assumes residuals are orthogonal across assets at each time slice, as they are in the independent per-asset generator. Real residuals share sector and style factors, which reduce $\text{Var}(\varepsilon_i - \varepsilon_j)$ below the independent-generator value $\sigma_{\varepsilon,i}^2 + \sigma_{\varepsilon,j}^2$. Matching each marginal variance does not restore this covariance. The synthetic

D_ε will therefore exceed its real-market counterpart when the omitted covariance is positive. Quantifying that contribution requires a joint-residual experiment with the marginals and the allocation rule held fixed, which we did not run. Realized diversification returns on real equity portfolios are also attenuated relative to the second-order formula at all rebalancing horizons (Bouchey et al., 2015), so the synthetic excess is an upper bound on what an independent-residual generator would contribute in a live comparison.

S3. Validation metric definitions

To distinguish marginal agreement from temporal agreement, we used separate distributional, tail, and autocorrelation summaries (Tables 2 and 6). For one observed series $x = (x_1, \dots, x_n)$ and R synthetic paths $y^{(r)}$ of the same length, each distributional comparison used one complete synthetic path and the fixed observed history. The two-sample Kolmogorov–Smirnov (KS) statistic is the largest absolute difference between their empirical cumulative distribution functions, $D = \sup_z |F_x(z) - F_{y^{(r)}}(z)|$ (Kolmogorov, 1933; Smirnov, 1948). The Anderson–Darling (AD) comparison gives greater weight to differences near the tails (Anderson and Darling, 1952). We used the approximate two-sample KS test and the modified, tie-aware two-sample AD calculation in `HypothesisTests.jl`, with asymptotic p -values. A path passed when $p > 0.05$, and the pass percentage was 100 times the fraction of passing paths. Dependence within returns, shared observed data, and parameter fitting prevent these percentages from being interpreted as calibrated significance levels or evidence of model equivalence. They describe agreement with the supplied history.

To quantify marginal discrepancies without a pass threshold, we computed Wasserstein-1 and histogram Hellinger scores for each path (Table 2). Let $x_{(j)}$ and $y_{(j)}^{(r)}$ denote the j th order statistics in increasing order. For equal sample lengths, the Wasserstein-1 distance is:

$$W_1(x, y^{(r)}) = \frac{1}{n} \sum_{j=1}^n |x_{(j)} - y_{(j)}^{(r)}|. \quad (\text{S14})$$

It has the growth-rate units yr^{-1} . For each Hellinger comparison, we formed 50 equally spaced bins between the minimum and maximum of the pooled pair of samples. Let p_b and q_b be the observed and simulated bin counts divided by their original sample lengths. The implemented histogram score is:

$$H(x, y^{(r)}) = \left[\frac{1}{2} \sum_{b=1}^{50} (\sqrt{p_b} - \sqrt{q_b})^2 \right]^{1/2}. \quad (\text{S15})$$

The stored implementation used lower-inclusive, upper-exclusive bins, including the final bin, so observations equal to the pooled maximum were excluded. The resulting bin weights can sum to less than one. The reported score retains that endpoint convention and depends on the pair-specific bin edges; it is a histogram

approximation. Smaller Wasserstein and Hellinger scores indicate closer marginal agreement.

To describe tail shape, we computed excess kurtosis for each synthetic path and compared its mean with the observed value (Table 2). With $m_p(x) = n^{-1} \sum_{t=1}^n (x_t - \bar{x})^p$, the estimator was $\kappa(x) = m_4(x)/m_2(x)^2 - 3$, without a finite-sample bias correction. The composition scorecard also used the Hill index of absolute returns (Table 3). Let $X_{(1)} \geq \dots \geq X_{(n)}$ be decreasingly ordered absolute returns and $k = \max\{2, \lfloor 0.05n \rfloor\}$. The implemented upper-tail estimator is:

$$\hat{\xi} = \frac{1}{k-1} \sum_{j=1}^{k-1} \log \left(\frac{X_{(j)}}{X_{(k)}} \right). \quad (\text{S16})$$

The threshold $X_{(k)}$ must be positive. Larger values indicate a heavier estimated absolute-return tail. These shape summaries do not measure tail-threshold coverage.

To assess whether the simulated distributions bracketed the observed one, we measured agreement at 99 quantile levels (Table 2). For $u_j = j/100$, $j = 1, \dots, 99$, let $Q_x(u_j)$ be the observed sample quantile and $Q_r(u_j)$ the corresponding quantile of path r . Let L_j and U_j be the 5th and 95th percentiles of $\{Q_r(u_j) : r = 1, \dots, R\}$. All quantiles used the default linear interpolation between order statistics (type 7). The reported quantile-envelope coverage is:

$$C = \frac{100}{99} \sum_{j=1}^{99} \mathbf{1}\{L_j \leq Q_x(u_j) \leq U_j\}. \quad (\text{S17})$$

Larger values mean that more observed quantiles fall inside the pointwise simulation envelopes. This statistic assesses marginal quantile agreement; it does not estimate daily Value-at-Risk exceedance coverage or simultaneous confidence coverage.

To quantify persistence in return magnitude, we used the sample autocorrelation function (ACF) of absolute returns. Each series was centered by its own full-series absolute-return mean, $\bar{a}_x = n^{-1} \sum_t |x_t|$. At lag ℓ , the calculation was:

$$a_\ell(x) = \frac{\sum_{t=1}^{n-\ell} (|x_t| - \bar{a}_x)(|x_{t+\ell}| - \bar{a}_x)}{\sum_{t=1}^n (|x_t| - \bar{a}_x)^2}. \quad (\text{S18})$$

The denominator used all n observations, with no lag-dependent rescaling. Lag zero was excluded from the error scores. The single-asset comparison measured the error of the ensemble-average ACF (Table 2). For a lag limit L , that mean absolute error is:

$$E_L^{\text{ensemble}} = \frac{1}{L} \sum_{\ell=1}^L \left| a_\ell(x) - \frac{1}{R} \sum_{r=1}^R a_\ell(y^{(r)}) \right|. \quad (\text{S19})$$

We used $R = 1,000$ and $L = \min(252, n-1)$: lags 1–252 for the 2,766-observation training series and 1–248 for the 249-observation holdout. The state-resolution and emission comparisons used the same ensemble convention.

The multi-asset jump comparison instead measured error within each path before averaging (Table 6). For one asset, its mean path error is:

$$E_L^{\text{path}} = \frac{1}{R} \sum_{r=1}^R \left[\frac{1}{L} \sum_{\ell=1}^L |a_\ell(x) - a_\ell(y^{(r)})| \right]. \quad (\text{S20})$$

We used $L = 25$ and 60 in both windows and averaged these scores over assets. The fallback control also used the path convention with $L = 25$ (Table S18). At a common lag limit and on the same paths, the triangle inequality gives $E_L^{\text{ensemble}} \leq E_L^{\text{path}}$: averaging the curves can cancel errors of opposite sign. The two scores therefore answer different questions about temporal agreement. Their magnitudes cannot be compared directly across the single-asset and composition experiments, which also differ in lag limits, histories, and assets.

To describe simulation uncertainty in the single-asset comparison, we used the sampling unit appropriate to each statistic (Table 2). For a pass fraction $\hat{p}$, the reported binomial standard error was $100\sqrt{\hat{p}(1-\hat{p})/R}$ percentage points. This estimate is zero when every path passes or fails; it does not establish that the underlying pass probability is exactly zero or one. For kurtosis, Wasserstein distance, and Hellinger score, the standard error was the sample standard deviation of the R path scores divided by $\sqrt{R}$. For ensemble ACF error and quantile-envelope coverage, we used 500 bootstrap resamples of the R path indices, each of size R with replacement. Each resample retained entire paths and recomputed the statistic across all lags or quantile levels. The standard error was the sample standard deviation of those 500 recomputed values, without a further division by $\sqrt{500}$. All these errors condition on one observed history and a fixed fitted model; they omit uncertainty in fitting, model selection, and market regimes. The multi-asset jump contrasts instead used the 1,000 replication-level means over the asset set, preserving the shared market draw when calculating paired standard errors (Table S12).

S3.1. Gated recurrent unit benchmark

To provide an autoregressive neural benchmark, we used a gated recurrent unit (GRU) network to predict the distribution of the next SPY growth rate from the preceding 50 values (Table 2). We standardized the training series using its mean and population standard deviation, both fixed for evaluation. The network had two GRU layers with 64 hidden units per layer, dropout probability 0.1 between layers during training, and separate linear heads for a conditional mean and log variance. The last hidden output of each input window supplied both heads, giving 37,954 trainable parameters. For a standardized target z_t and head outputs m_t and v_t , the training loss was the Gaussian negative log-likelihood with its constant term omitted:

$$\mathcal{L} = \frac{1}{2B} \sum_{t \in \mathcal{B}} [v_t + (z_t - m_t)^2 \exp(-v_t)], \quad (\text{S21})$$

where $\mathcal{B}$ is a training batch containing $B = 256$ windows. The 2,766 training observations supplied 2,716 overlapping input-target pairs. Each epoch shuffled

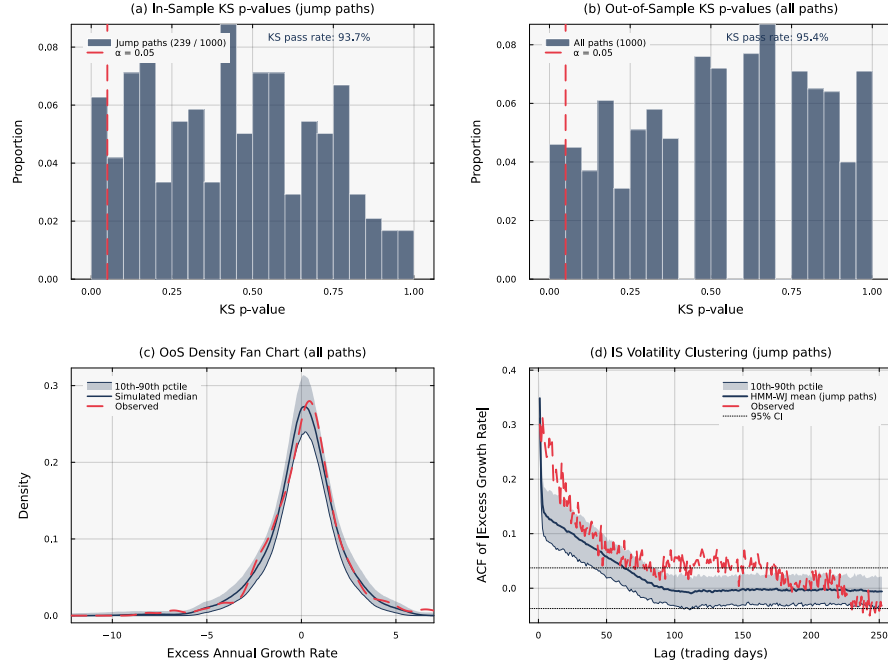


Figure S2: **Statistical validation of the hidden Markov model with jumps (HMM-WJ)** ($N = 100$, $\alpha = 0.05$). Panels (a) and (d) condition on the $\sim 24\%$ of in-sample paths containing jump events. (a) In-sample KS p -values for 239 jump paths (pass rate 93.7%). (b) Out-of-sample KS p -values for all 1,000 paths (pass rate 95.4%). (c) Out-of-sample density fan chart; the observed density falls within the simulation envelope. (d) Observed and mean simulated in-sample autocorrelation function (ACF) of $|G_t|$ for jump paths.

Table S1: **Descriptive statistics and stylized-facts tests for SPY daily excess growth rates.** In-sample (IS): 2014–2024 ($T = 2,766$); out-of-sample (OoS): 2025 ($T = 249$). Gaussian and Laplace columns show maximum likelihood estimation (MLE) fits to the in-sample data. We used the Jarque–Bera (JB) normality test and the Ljung–Box (LB) autocorrelation test at lag 20. * denotes rejection at $\alpha = 0.05$.

Statistic	IS Observed	OoS Observed	Gaussian	Laplace
Mean (annualized, %)	6.31	11.60	6.31	14.19
Standard deviation of G_t (% yr $^{-1}$)	214.50	220.62	214.46	205.93
Skewness	−0.753	−1.093	0	0
Excess Kurtosis	7.715	6.867	0	3
JB normality test	reject* ($p < 0.001$)	reject* ($p < 0.001$)	not used	not used
LB test on G_t (lag 20)	reject* ($p < 0.001$)	reject* ($p = 0.019$)	not used	not used
LB test on $ G_t $ (lag 20)	reject* ($p < 0.001$)	reject* ($p < 0.001$)	not used	not used

these pairs and used ten complete batches; the final incomplete batch was dropped. Adam used an initial learning rate of 10^{-3} , default moment decay rates (0.9, 0.999), and no weight decay. We clipped the gradient norm at one and halved the learning rate after the scheduler’s 15-epoch plateau tolerance, with a floor of 10^{-5} . Training ran for at most 200 epochs. We retained weights whenever mean training loss improved by more than 10^{-4} and stopped after 40 epochs without such improvement. The recorded fit completed 200 epochs and retained epoch 199. Model selection used training loss, with no separate validation split; the holdout supplied only the evaluation horizon.

To generate paths, we disabled dropout and selected an initial window uniformly from the 2,716 training input windows. At each step, we passed the current 50 values through the network with a fresh zero hidden state, sampled a Gaussian value with mean m_t and standard deviation $\max\{\exp(v_t/2), 10^{-6}\}$, and replaced the oldest window value with that draw. Every generated value was retained, with no burn-in, and the fixed training normalization was inverted afterward. The same fitted network generated 1,000 training-length and then 1,000 holdout-length paths, in that order, using seed 1234 for NumPy and PyTorch. The recorded rerun used Python 3.14.2, PyTorch 2.10.0, NumPy 2.4.3, and SciPy 1.17.1 on a central processing unit (CPU). The repository includes pinned dependencies, trained weights, an epoch log, and input and output fingerprints. Parallel generation assigned each worker the random-number state for its position in the original serial rollout; a direct check confirmed identical paths. The rerun reproduced both archived path files exactly. The benchmark is one fitted network: its reported simulation errors do not measure variation across training seeds or neural architectures.

S4. Derivation of the single-asset generator

This appendix derives the probability law of the fitted single-asset generator and connects each fitted component to the empirical feature it controls. The hidden Markov model with jumps (HMM-WJ), its no-jump restriction (HMM-NJ), and their symbols follow the main-text definitions in Eq. (2) and Algorithm 1.

Unless noted otherwise, numerical values evaluate the closed-form expressions at the fitted SPY model of Table 1: $N = 100$ states, Student- t degrees of freedom $\nu = 5$, jump probability $\epsilon = 10^{-4}$, mean episode duration $\lambda = 90$ steps, tail-selection probability $p_{\text{neg}} = 0.52$, tail-set size $N_{\text{tail}} = 5$, and the in-sample series of $T = 2,766$ daily excess growth rates. Supplementary Table S2 collects these values beside their sample or simulated counterparts.

Table S2: **Closed-form quantities of the fitted SPY generator against sample and simulated values.** Each closed-form entry evaluates the expression named in the text at the fitted SPY model of Table 1. Sample values are computed from the in-sample SPY series; simulated values use 400 in-sample-length paths for the jump-episode quantities and 1,000 paths for skewness. For comparison with the sup-distance entry, the two-sample Kolmogorov–Smirnov critical value at $\alpha = 0.05$ for two samples of $T = 2,766$ observations is $1.358\sqrt{2/T} = 0.0365$. Observed autocorrelations are those of the in-sample series shown in Fig. 1d.

Quantity	Closed form at the fit	Sample or simulation
<i>Stationary law and moments</i>		
State occupancy count, smallest and largest (of $T = 2,766$)	–	16, 47
Occupancy frequency $\hat{\pi}_k$, smallest and largest	–	0.0058, 0.017
$\max_k \hat{\pi}_k - \bar{\pi}_k $	3.7×10^{-4}	–
Mean (yr^{-1})	0.0636	0.0631
$\sup_x F_{\text{mix}}(x) - \hat{F}_T(x) $	0.0045	–
Within-state variance V_w (yr^{-2}), share of sample variance	0.17, 3.6%	–
Variance (yr^{-2}), Student- t emissions	4.72	4.60
Variance (yr^{-2}), Gaussian emissions	4.61	4.60
Standard deviation (yr^{-1}), Student- t emissions	2.17	2.14
Excess kurtosis, Student- t emissions	8.34	7.72
Excess kurtosis, Gaussian emissions	5.82	7.72
Skewness, HMM-NJ (1,000 simulated paths)	–	–0.76
Skewness, HMM-WJ (1,000 simulated paths)	–	–0.73
<i>Dependence without jumps</i>		
$ \theta_2 $, smallest lag with $ \theta_2 ^r < 0.01$	0.253, lag 4	–
ACF of $ G_t $ at lag 1, no jumps	0.26	0.30 (observed)
ACF of $ G_t $ at lag 3, no jumps	0.021	0.31 (observed)
ACF of $ G_t $ at lag 5, no jumps	0.001	0.26 (observed)
ACF of $ G_t $ at lag 25, no jumps	< 0.001	0.14 (observed)
<i>Jump-episode arithmetic</i>		
Forced-step fraction π_J	0.0089	0.0088 (400 paths)
Jump-path probability	24.2%	24.8% (99 of 400)
Forced fraction within a jump path	3.3% (λ/T)	3.6%
Forced-step standard deviation (yr^{-1})	5.38	–
Mean absolute return, forced $\bar{a}_J$ and free $\bar{a}$ (yr^{-1})	4.96, 1.47	–
$\text{Var}(A_t)$, single-episode conditioning	3.01	–
Predicted jump-path ACF plateau	0.13	–
Excess kurtosis, single-episode jump path	7.20	–
Variance inflation from forced steps, single-episode jump path	17%	–

S4.1. Generative law

The no-jump model generates a hidden state path $S_0, \dots, S_T$ on $\{1, \dots, N\}$ and observations $G_1, \dots, G_T$ with the joint density:

$$p(s_0, \dots, s_T, g_1, \dots, g_T) = \bar{\pi}_{s_0} \prod_{t=1}^T \mathbf{T}_{s_{t-1}, s_t} f_{s_t}(g_t), \quad (\text{S22})$$

where $\bar{\pi}_{s_0}$ is the initial probability of state s_0 , $\mathbf{T}_{s_{t-1}, s_t}$ is the probability of moving from state s_{t-1} to state s_t , and f_k is the emission density of state k .

Under Eq. (2), f_k is the location-scale Student- t density:

$$f_k(g) = \frac{1}{\sigma_k} \psi_\nu \left(\frac{g - \mu_k}{\sigma_k} \right), \quad \psi_\nu(x) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi} \Gamma(\frac{\nu}{2})} \left(1 + \frac{x^2}{\nu} \right)^{-\frac{\nu+1}{2}}, \quad (\text{S23})$$

where ψ_ν is the density of a standard Student- t random variable with ν degrees of freedom, Γ is the gamma function, and (μ_k, σ_k) are the location and scale of state k . The factorization in Eq. (S22) states the structural property that drives everything that follows: given the state path, the observations are independent. Every form of serial dependence the generator can produce must therefore be carried by the state path (Rabiner and Juang, 1986).

The jump-augmented model enlarges the hidden state. Let $J_t \in \{0, 1, 2, \dots\}$ count the forced steps remaining after time t , and let u be the forced-step state distribution, which mixes uniform draws over the two tail sets:

$$u_k = \begin{cases} p_{\text{neg}}/N_{\text{tail}} & k \in \mathcal{S}_-, \\ (1 - p_{\text{neg}})/N_{\text{tail}} & k \in \mathcal{S}_+, \\ 0 & \text{otherwise.} \end{cases} \quad (\text{S24})$$

Writing $p_K(m) = e^{-\lambda} \lambda^m / m!$ for the Poisson probability of episode length m and $\mathbf{1}\{\cdot\}$ for the indicator function, the pair (S_t, J_t) evolves by the transition kernel:

$$\begin{aligned} & \Pr(S_t = k, J_t = j \mid S_{t-1} = s, J_{t-1} = i) \\ &= \begin{cases} u_k \mathbf{1}\{j = i - 1\} & i \geq 1, \\ (1 - \epsilon + \epsilon e^{-\lambda}) \mathbf{T}_{s,k} \mathbf{1}\{j = 0\} + \epsilon p_K(j + 1) u_k & i = 0, \end{cases} \end{aligned} \quad (\text{S25})$$

and the emission draw depends only on the first coordinate, $G_t \sim f_{S_t}$. The two cases read as follows. While an episode is active ($i \geq 1$), the next state is an independent draw from u and the counter decrements; the tail set and the state within it are re-drawn at every forced step. While no episode is active ($i = 0$), the chain either takes an ordinary transition (no trigger, or a trigger whose Poisson draw returns zero) or starts an episode of length $j + 1 \geq 1$ whose first forced step occurs immediately. The pair (S_t, J_t) is therefore itself a Markov chain, and the HMM-WJ generator is a hidden Markov model on the enlarged state space $\{1, \dots, N\} \times \{0, 1, 2, \dots\}$. Algorithm 1 implements Eq. (S25) with the episode counter truncated at the simulation horizon; it emits the stationary initial draw as the first observation and applies the kernel from the second step onward, which changes the law of a path only through the absent possibility of a trigger at the first step. When an episode ends, ordinary transitions resume from the last forced tail state, so the chain re-enters the body of the state space through the tail rows of $\mathbf{T}$.

S4.2. Estimators and the one-step empirical law

The fitting procedure consists of four closed-form estimators, and each one matches a specific empirical object exactly. The Laplace partition comes first.

For observations $G_1, \dots, G_T$, the Laplace log-likelihood with location μ_L and scale b_L is:

$$\ell(\mu_L, b_L) = -T \ln(2b_L) - \frac{1}{b_L} \sum_{t=1}^T |G_t - \mu_L|. \quad (\text{S26})$$

For any fixed $b_L > 0$, maximizing ℓ over μ_L minimizes $\sum_t |G_t - \mu_L|$, so $\hat{\mu}_L$ is the sample median. Setting the b_L derivative to zero at $\hat{\mu}_L$ gives:

$$\begin{aligned} \frac{\partial \ell}{\partial b_L} &= -\frac{T}{b_L} + \frac{1}{b_L^2} \sum_{t=1}^T |G_t - \hat{\mu}_L| = 0 \quad (\text{first-order condition}) \\ \hat{b}_L &= \frac{1}{T} \sum_{t=1}^T |G_t - \hat{\mu}_L|, \quad (\text{solve for } b_L) \end{aligned} \quad (\text{S27})$$

the mean absolute deviation about the median. The state boundaries then invert the fitted Laplace cumulative distribution function (CDF) (Kotz et al., 2001). The Laplace CDF is piecewise exponential:

$$F_L(x) = \begin{cases} \frac{1}{2} \exp((x - \hat{\mu}_L)/\hat{b}_L) & x \leq \hat{\mu}_L, \\ 1 - \frac{1}{2} \exp(-(x - \hat{\mu}_L)/\hat{b}_L) & x > \hat{\mu}_L, \end{cases} \quad (\text{S28})$$

and setting $F_L(Q_k) = k/N$ and solving each branch for Q_k gives the closed-form boundaries:

$$\begin{aligned} \frac{k}{N} &= \frac{1}{2} e^{(Q_k - \hat{\mu}_L)/\hat{b}_L} \implies Q_k = \hat{\mu}_L + \hat{b}_L \ln \frac{2k}{N}, \quad (\text{lower branch}) \\ \frac{k}{N} &= 1 - \frac{1}{2} e^{-(Q_k - \hat{\mu}_L)/\hat{b}_L} \implies Q_k = \hat{\mu}_L - \hat{b}_L \ln \left(2 - \frac{2k}{N}\right). \quad (\text{upper branch}) \end{aligned} \quad (\text{S29})$$

The bins are equiprobable under the fitted Laplace, not under the empirical distribution, so the in-sample state counts are not uniform: across the 100 states, they range from 16 to 47 of the 2,766 observations (Supplementary Table S2). Every within-state mean μ_k and standard deviation σ_k was therefore computed from at least 16 observations, and no state required a fallback to the global moments.

The transition matrix is the maximum likelihood estimator (MLE) of a first-order Markov chain. Conditional on the assigned state sequence $s_1, \dots, s_T$, let n_{jk} count the transitions from state j to state k and let $n_{j\cdot} = \sum_k n_{jk}$ be the row total. The likelihood of the transition parameters factorizes over rows into multinomials, and maximizing the log-likelihood subject to each row summing to one, with multiplier c_j for row j , uses the Lagrangian:

$$\mathcal{L} = \sum_{j,k} n_{jk} \ln \mathbf{T}_{jk} + \sum_j c_j \left(1 - \sum_k \mathbf{T}_{jk}\right), \quad (\text{S30})$$

whose stationarity conditions solve in three steps:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial \mathbf{T}_{jk}} &= \frac{n_{jk}}{\mathbf{T}_{jk}} - c_j = 0 && \text{(first-order condition)} \\
\mathbf{T}_{jk} &= \frac{n_{jk}}{c_j} && \text{(solve for } \mathbf{T}_{jk} \text{)} \\
c_j &= \sum_k n_{jk} = n_{j\cdot} && \text{(row constraint } \sum_k \mathbf{T}_{jk} = 1 \text{)} \\
\hat{\mathbf{T}}_{jk} &= \frac{n_{jk}}{n_{j\cdot}} && \text{(counted estimator)} \tag{S31}
\end{aligned}$$

For a row with $n_{j\cdot} > 0$, the interior condition applies to cells with positive counts; for a cell with $n_{jk} = 0$, the log-likelihood is non-increasing in $\mathbf{T}_{jk}$, so its maximizer is the boundary value $\hat{\mathbf{T}}_{jk} = 0$, which the same formula returns. When $n_{j\cdot} = 0$, the observations do not identify that row. The implementation sets every entry to $1/N$; this is a fallback choice, not an estimate of an observed transition frequency. Multiplying $\hat{\mathbf{T}}_{jk}$ by the row frequency $n_{j\cdot}/(T-1)$ recovers $n_{jk}/(T-1)$, the empirical frequency of the state pair (j, k) . The counted estimator therefore reproduces the in-sample joint law of (S_{t-1}, S_t) exactly; everything it asserts about longer horizons is the Chapman–Kolmogorov extension $\mathbf{T}^\tau$ of that one-step law (Bilmes, 1998; Grinstead and Snell, 1997). The emission estimator completes the pair: within each state with at least two observations and sample standard deviation at least 10^{-12} yr^{-1} , μ_k and σ_k are the sample mean and standard deviation of the assigned observations. Otherwise, the implementation uses the fallback rules in Eq. (S53). In either case, σ_k enters Eq. (S23) as the scale, so the emission standard deviation is $\sigma_k \sqrt{\nu/(\nu-2)}$. The aggregate consequence of this scale convention is quantified in Eq. (S35).

The initial distribution ties the pieces together. Let n_k count the occupancy of state k over the T observations, and let $\hat{\pi}_k = n_k/T$ be the occupancy frequency. The row total satisfies $n_{j\cdot} = n_j - \mathbf{1}\{s_T = j\}$, and the column total satisfies $n_{\cdot k} = n_k - \mathbf{1}\{s_1 = k\}$, because only the last observation opens no transition and only the first closes none. Substituting both identities gives:

$$\begin{aligned}
(\hat{\pi} \hat{\mathbf{T}})_k &= \sum_j \frac{n_j}{T} \cdot \frac{n_{jk}}{n_{j\cdot}} && \text{(definitions)} \\
&= \sum_j \frac{n_{j\cdot}}{T} \cdot \frac{n_{jk}}{n_{j\cdot}} + \sum_j \frac{\mathbf{1}\{s_T = j\}}{T} \cdot \frac{n_{jk}}{n_{j\cdot}} && (n_j = n_{j\cdot} + \mathbf{1}\{s_T = j\}) \\
&= \frac{n_{\cdot k}}{T} + \frac{\hat{\mathbf{T}}_{s_T, k}}{T} && \text{(sum transitions into } k \text{)} \\
&= \hat{\pi}_k + \frac{\hat{\mathbf{T}}_{s_T, k} - \mathbf{1}\{s_1 = k\}}{T} && (n_{\cdot k} = n_k - \mathbf{1}\{s_1 = k\}) \tag{S32}
\end{aligned}$$

The occupancy frequencies therefore satisfy the stationarity equation $\hat{\pi} \hat{\mathbf{T}} = \hat{\pi}$ up to a fixed-point residual bounded by $1/T$. For the fitted SPY model, the solved

eigenvector $\bar{\pi}$ deviates from the occupancy frequencies by at most 3.7×10^{-4} , the same order as $1/T$, against occupancy values between 0.0058 and 0.017 (Supplementary Table S2). The initial distribution is consequently not an abstract spectral object: $\bar{\pi}$ is, up to a correction of the order of $1/T$, the vector of in-sample state frequencies.

S4.3. Stationary distribution and moments

The stationary observation law of the no-jump generator follows directly from Eq. (S22). Marginalizing the state at a single time gives the mixture density and CDF:

$$p(g) = \sum_{k=1}^N \bar{\pi}_k f_k(g), \quad F_{\text{mix}}(x) = \sum_{k=1}^N \bar{\pi}_k \Psi_{\nu} \left(\frac{x - \mu_k}{\sigma_k} \right), \quad (\text{S33})$$

where Ψ_{ν} is the CDF of the standard Student- t with ν degrees of freedom. Every ingredient of Eq. (S33) is a summary of the fitted SPY model. Its component locations and scales are within-state sample moments because no fallback was needed; its stationary weights differ from occupancy frequencies through the end effects described above. For other assets, fallback locations or scales need not be within-state estimates (Table S17). Mean matching therefore requires a qualification: weighting empirical state means by the occupancy frequencies telescopes into the grand mean, whereas replacing a populated state's mean with a global fallback need not preserve that identity. With empirical state means, the calculation is:

$$\begin{aligned} \sum_{k:n_k>0} \hat{\pi}_k \mu_k &= \sum_{k:n_k>0} \frac{n_k}{T} \cdot \frac{1}{n_k} \sum_{t:s_t=k} G_t \quad (\text{definitions of } \hat{\pi}_k, \mu_k) \\ &= \frac{1}{T} \sum_{t=1}^T G_t, \quad (\text{each } t \text{ counted once}) \end{aligned} \quad (\text{S34})$$

and replacing $\hat{\pi}$ by $\bar{\pi}$ changes the value only through the $1/T$ correction of Eq. (S32): the model mean is 0.0636 yr^{-1} against the sample mean of 0.0631 yr^{-1} (Supplementary Table S2). The distributional match is equally direct. Evaluated at the fitted SPY model, the largest gap between the model CDF and the empirical CDF is $\sup_x |F_{\text{mix}}(x) - \hat{F}_T(x)| = 0.0045$ (Supplementary Table S2), where $\hat{F}_T$ is the in-sample empirical CDF. The two-sample Kolmogorov–Smirnov (KS) critical value at significance level $\alpha = 0.05$ for two samples of 2,766 observations is $1.358\sqrt{2/T} = 0.0365$, so the model-versus-data distance consumes about one eighth of the rejection threshold. Rejections of simulated paths are therefore driven by path-level sampling variation rather than by a misplaced marginal, which is consistent with the in-sample pass rates in Supplementary Table S4.

The second moment makes the role of the emission scale convention explicit.

Conditioning on the state and applying the law of total variance gives:

$$\begin{aligned}\text{Var}(G_t) &= \mathbb{E}[\text{Var}(G_t | S_t)] + \text{Var}(\mathbb{E}[G_t | S_t]) \quad (\text{law of total variance}) \\ &= \frac{\nu}{\nu-2} \sum_k \bar{\pi}_k \sigma_k^2 + \sum_k \bar{\pi}_k (\mu_k - \bar{\mu})^2, \quad (\text{Student-}t \text{ variance}) \quad (\text{S35})\end{aligned}$$

where $\bar{\mu} = \sum_k \bar{\pi}_k \mu_k$ is the stationary mean. Define the within-state variance $V_w = \sum_k \bar{\pi}_k \sigma_k^2$ and the between-state variance $V_b = \sum_k \bar{\pi}_k (\mu_k - \bar{\mu})^2$. The corresponding empirical decomposition of the sample variance is $V_b + V_w$, up to degrees-of-freedom corrections, so the model variance exceeds the sample variance by $2V_w/(\nu-2)$. At the SPY fit (Supplementary Table S2), $V_w = 0.17 \text{ yr}^{-2}$, only 3.6% of the total variance of 4.60 yr^{-2} : the quantile partition concentrates almost all dispersion between states rather than within them. The inflation is correspondingly small, a model variance of 4.72 against 4.60 yr^{-2} (2.17 against 2.14 yr^{-1} in standard deviation), and vanishes under Gaussian emissions (4.61 yr^{-2}).

The fourth moment explains the emission comparison in Supplementary Table S4. Write $G_t = \mu_k + \sigma_k X$ within state k , where X is a standard Student- t draw, and let $\delta_k = \mu_k - \bar{\mu}$. Expanding the fourth central moment and dropping the odd-power terms, which vanish by the symmetry of ψ_ν , gives:

$$\begin{aligned}\mathbb{E}[(G_t - \bar{\mu})^4] &= \sum_k \bar{\pi}_k \mathbb{E}[(\delta_k + \sigma_k X)^4] \quad (\text{condition on the state}) \\ &= \sum_k \bar{\pi}_k [\delta_k^4 + 6\delta_k^2 \sigma_k^2 \mathbb{E}(X^2) \\ &\quad + \sigma_k^4 \mathbb{E}(X^4)],\end{aligned} \quad (\text{S36})$$

with the standard Student- t moments, finite for $\nu > 4$:

$$\mathbb{E}(X^2) = \frac{\nu}{\nu-2}, \quad \mathbb{E}(X^4) = \frac{3\nu^2}{(\nu-2)(\nu-4)}. \quad (\text{S37})$$

At $\nu = 5$ these values are $5/3$ and 25 : the within-state kurtosis is 9 , against 3 for Gaussian emissions. The excess kurtosis of the stationary mixture is $\mathbb{E}[(G_t - \bar{\mu})^4]/\text{Var}(G_t)^2 - 3$; evaluating it at the SPY fit gives 8.34 with Student- t emissions and 5.82 with Gaussian emissions, against an observed value of 7.72 (Supplementary Table S2). The emission choice enters Eq. (S36) through both moments in Eq. (S37); the larger change by far is $\mathbb{E}(X^4)$ rising from 3 to 25 , which acts on the σ_k^4 terms that the wide tail states dominate. The per-path simulated values reported in Supplementary Table S4 (7.5 and 5.5) sit below these population values because the sample kurtosis of a heavy-tailed series of $2,766$ observations is biased downward, but the population gap of 2.5 between the emission choices matches the simulated gap of 2.0 in both direction and size. The mixture also carries the asymmetry of the fitted state occupancies without an explicit skewness parameter: simulating $1,000$ in-sample-length paths gives a mean sample skewness of -0.76 for HMM-NJ and -0.73 for HMM-WJ, against

the observed -0.75 (Supplementary Tables S2 and S1). Because the states partition the Laplace quantiles of an asymmetric series, the occupancy-weighted mixture inherits that asymmetry through the μ_k and σ_k estimates.

S4.4. Dependence structure without jumps

The results stated so far concern single time slices; the dependence structure is where the no-jump generator falls short of the data, and the failure is structural rather than a matter of estimation. Within state k , the sojourn time is geometric: the chain remains for m steps with probability $\mathbf{T}_{kk}^{m-1}(1 - \mathbf{T}_{kk})$ and therefore stays for $1/(1 - \mathbf{T}_{kk})$ steps on average, between one and two days for every fitted state (Supplementary Fig. S3c). The autocorrelation of absolute returns inherits this short memory. Let $a_k = \mathbb{E}[|G_t| \mid S_t = k]$ be the within-state mean absolute return and $\bar{a} = \sum_k \bar{\pi}_k a_k$ its stationary average. Because observations are conditionally independent given the states, the lag- τ product moment reduces to a state-pair sum:

$$\begin{aligned} \mathbb{E}[|G_t| |G_{t+\tau}|] &= \sum_{j,k} \Pr(S_t = j, S_{t+\tau} = k) a_j a_k \quad (\text{conditional independence}) \\ &= \sum_{j,k} \bar{\pi}_j (\mathbf{T}^\tau)_{jk} a_j a_k, \quad (\text{Chapman-Kolmogorov}) \end{aligned} \tag{S38}$$

so the autocorrelation function (ACF) of $|G_t|$ is:

$$\text{ACF}_{|G|}(\tau) = \frac{\sum_{j,k} \bar{\pi}_j (\mathbf{T}^\tau)_{jk} a_j a_k - \bar{a}^2}{\text{Var}(|G_t|)}. \tag{S39}$$

As τ grows, $\mathbf{T}^\tau$ converges to its rank-one limit, every row equal to $\bar{\pi}$, at the geometric rate $|\theta_2|^\tau$, where θ_2 is the second-largest eigenvalue of $\mathbf{T}$ in modulus; the ACF is therefore bounded by a constant multiple of $|\theta_2|^\tau$. For the fitted SPY matrix (Supplementary Table S2), $|\theta_2| = 0.253$, and $|\theta_2|^\tau$ falls below 0.01 from lag 4 onward. Evaluating Eq. (S38) gives a model ACF of 0.26 at lag 1, close to the observed 0.30 because the counted matrix pins the one-step state-pair law, but 0.021 at lag 3 and effectively zero from lag 5, while the observed ACF remains near 0.26 at lag 5 and 0.14 at lag 25 (Supplementary Table S2; Fig. 1d). No first-order chain fitted by transition counting can do otherwise: matching the one-step law fixes $\mathbf{T}$, and the spectral gap of that matrix then forces geometric mixing within a week. This is the quantitative content of the main-text statement that the counted transition probabilities leave extreme-return states too quickly to reproduce volatility clustering, and it is the failure the jump mechanism corrects.

S4.5. Jump-episode arithmetic

The jump mechanism overlays the fast-mixing chain with rare, long episodes, and its operating characteristics follow from renewal arguments. On each step

with no active episode, an episode with at least one forced step begins with probability $p_e = \epsilon(1 - e^{-\lambda})$, since a trigger whose Poisson draw returns zero is discarded; conditional on starting, the episode length is the zero-truncated Poisson draw with mean $\mathbb{E}[K \mid K \geq 1] = \lambda/(1 - e^{-\lambda})$. At $\lambda = 90$, the correction $e^{-\lambda}$ is below 10^{-39} , so $p_e = \epsilon$ and $\mathbb{E}[K \mid K \geq 1] = \lambda$ to machine precision. The long-run fraction of forced steps, denoted π_J , follows from the renewal-reward theorem. A renewal cycle consists of the free steps preceding an episode and the episode itself: the number of free trials up to and including the trigger is geometric with mean $1/p_e$, the triggering trial is itself the first forced step, and the episode contributes $\mathbb{E}[K \mid K \geq 1]$ forced steps in total:

$$\begin{aligned}
\pi_J &= \frac{\mathbb{E}[K \mid K \geq 1]}{\frac{1 - p_e}{p_e} + \mathbb{E}[K \mid K \geq 1]} && \text{(forced steps over cycle length)} \\
&= \frac{p_e \mathbb{E}[K \mid K \geq 1]}{1 - p_e + p_e \mathbb{E}[K \mid K \geq 1]} && \text{(multiply through by } p_e) \\
&= \frac{\epsilon\lambda}{1 - \epsilon + \epsilon e^{-\lambda} + \epsilon\lambda} && (p_e \mathbb{E}[K \mid K \geq 1] = \epsilon\lambda) \\
&\approx \frac{\epsilon\lambda}{1 + \epsilon\lambda} = 0.0089. && \text{(drop } \epsilon e^{-\lambda}, \text{ then } \epsilon) \tag{S40}
\end{aligned}$$

Fewer than one step in a hundred is forced at the fitted values; the measured fraction across 400 simulated in-sample-length paths is 0.0088 (Supplementary Table S2). The product $\epsilon\lambda$ is the quantity that sets this fraction, a point that matters for identification in Section S4.6.

At the path level, the same arithmetic determines how episodes are distributed across the ensemble. A simulated path contains no episode only if every one of its $T - 1$ transition steps fails to trigger one, so the jump-path probability at $T = 2,766$ is $1 - (1 - p_e)^{T-1} = 24.2\%$; the simulated ensemble gives 24.8% (99 of 400 paths; Supplementary Table S2), matching the $\sim 24\%$ jump-path fraction conditioned on in Supplementary Fig. S2. Episodes begin at the long-run rate of one per expected renewal cycle, so a path carries about 0.274 episodes in expectation. The expected gap between episodes is $1/\epsilon = 10^4$ trading days, about 40 trading years, so the ensemble is a two-population mixture: roughly three quarters of the paths never trigger an episode and follow the HMM-NJ law exactly, while the remaining quarter carry a single episode (about one jump path in eight carries a second). Within a jump path, the forced fraction is $\lambda/T = 3.3\%$ for a single episode; the simulated mean is 3.6% (Supplementary Table S2), slightly higher through the multi-episode paths. Forced steps differ sharply from free ones. Under the forced-step distribution u of Eq. (S24), the conditional standard deviation is 5.38 yr^{-1} against the free-step 2.17 yr^{-1} , a variance ratio of 6.1, and the mean absolute return is $\bar{a}_J = \sum_k u_k a_k = 4.96 \text{ yr}^{-1}$ against $\bar{a} = 1.47 \text{ yr}^{-1}$ (Supplementary Table S2). The stationary state marginal of the augmented chain is the mixture $(1 - \pi_J)$ times the free-phase distribution plus π_J times u ; the free phase relaxes from the post-episode tail state back to $\bar{\pi}$ at the rate $|\theta_2|$ within a few steps. Using $\bar{\pi}$ as the initial law therefore

misallocates $O(\pi_J) \approx 1\%$ of probability mass, which quantifies the main-text remark that $\bar{\pi}$ is exact for the ordinary transition matrix but approximate for the jump-augmented process.

S4.6. Autocorrelation and kurtosis under jumps

The tuning objective of Eq. (3) scores the absolute-return ACF and the excess kurtosis, and both statistics admit closed-form approximations that show what (ϵ, λ) control. The ACF contribution rests on one more renewal quantity: the probability $h(\tau)$ that, given a step is forced, the step τ later belongs to the same episode. A uniformly chosen forced step lands in an episode with the length-biased law, an episode of length m covering m forced steps, and its position within that episode is uniform. Combining the two gives:

$$\begin{aligned}
 h(\tau) &= \sum_{m \geq 1} \frac{m \Pr(K = m \mid K \geq 1)}{\mathbb{E}[K \mid K \geq 1]} \cdot \frac{(m - \tau)_+}{m} \\
 &= \frac{\mathbb{E}[(K - \tau)_+ \mid K \geq 1]}{\mathbb{E}[K \mid K \geq 1]} && \text{(sum over } m) \\
 &\approx \left(1 - \frac{\tau}{\lambda}\right)_+, && (K \text{ concentrated near } \lambda)
 \end{aligned} \tag{S41}$$

where $(x)_+ = \max(x, 0)$ and the approximation holds because the Poisson length concentrates within a few multiples of $\sqrt{\lambda} \approx 9.5$ around λ . The same-episode probability therefore decays almost linearly from one to zero over the horizon λ . The ACF follows by conditioning on the forced-step indicator $Z_t = \mathbf{1}\{\text{step } t \text{ is forced}\}$. Given the indicator path, every forced step's state is an independent draw from u , so two forced observations are conditionally independent; the dependence linking a forced observation to a later free one runs only through the episode's final state and, like the free-phase covariance itself, decays at the $|\theta_2|^\tau$ rate of Section S4.4. For lags beyond that scale, only the conditional means covary. Write $A_t = |G_t|$ for the absolute return. The conditional mean is $\mathbb{E}[A_t \mid Z_t] = \bar{a} + Z_t(\bar{a}_J - \bar{a})$, and for rare episodes the joint forced probability is $\Pr(Z_t = 1, Z_{t+\tau} = 1) \approx \pi_J h(\tau)$, so:

$$\begin{aligned}
 \text{Cov}(A_t, A_{t+\tau}) &\approx \text{Cov}(\mathbb{E}[A_t \mid Z_t], \mathbb{E}[A_{t+\tau} \mid Z_{t+\tau}]) && \text{(independence given } Z) \\
 &= (\bar{a}_J - \bar{a})^2 \text{Cov}(Z_t, Z_{t+\tau}) && \text{(conditional mean in } Z) \\
 &\approx (\bar{a}_J - \bar{a})^2 \pi_J h(\tau), && \text{(episodes rare)} \tag{S42}
 \end{aligned}$$

so the jump mechanism adds an ACF component of height $\pi_J(\bar{a}_J - \bar{a})^2 / \text{Var}(A_t)$ that decays almost linearly to zero at lag λ , precisely the slow component the counted chain cannot produce. For a jump path carrying a single episode, the relevant forced fraction is λ/T rather than π_J , and evaluating the resulting expression at the fitted values, with $\text{Var}(A_t) = 3.01$ under the same conditioning, gives a predicted plateau of 0.13 (Supplementary Table S2). Supplementary Table S3 compares this prediction with the mean simulated jump-path ACF: the

Table S3: **Episode-overlap prediction of the jump-path absolute-return autocorrelation function (ACF) against simulation.** The prediction evaluates Eq. (S42) at the fitted SPY values with the single-episode forced fraction λ/T in place of π_J and the same-episode probability $h(\tau)$ of Eq. (S41). The simulated column is the mean ACF of $|G_t|$ over the 99 jump-active paths among 400 simulated in-sample-length paths. The free-phase component of Eq. (S38) is excluded from the prediction; it has decayed below 0.01 from lag 5 onward.

Lag τ	$h(\tau)$	Predicted ACF	Simulated ACF
1	0.989	0.130	0.348
2	0.978	0.129	0.199
3	0.967	0.127	0.153
5	0.944	0.124	0.137
10	0.889	0.117	0.123
15	0.833	0.110	0.115
20	0.778	0.102	0.105
25	0.722	0.095	0.099

two differ by 0.013 at lag 5 and by less than 0.01 from lag 10 onward. At lags 1 to 3 the simulated values exceed the episode component (0.35 against 0.13 at lag 1; Supplementary Table S3) because the free-phase geometric component of Section S4.4 adds on top before it dies out.

The kurtosis contribution pulls in the opposite direction. Treating a jump path as a two-regime mixture with weights $(1 - \lambda/T, \lambda/T)$ and evaluating the fourth-moment expression of Eq. (S36) with the forced-step distribution u in place of $\bar{\pi}$ for the episode regime gives a population excess kurtosis of 7.20 for a single-episode jump path, against 8.34 with no jumps (Supplementary Table S2). Forced steps inflate the variance, by about 17% at the fitted values, faster than they inflate the fourth moment, because u draws uniformly within each tail set rather than extending the extreme tail. Adding jump content therefore lowers kurtosis toward and past the observed value while it raises the slow ACF component; the jump-share sweep reported in the main text traces the same tradeoff empirically, with the Hill tail index flat across the sweep because the tail exponent is set by ν , not by the jump content. The objective of Eq. (3) balances these two effects.

The same expressions show how the two jump parameters are identified, and where they are not. Within a single-episode jump path, the ACF component of Eq. (S42) has height proportional to λ/T and horizon λ , and the kurtosis shift depends on the same forced fraction: every statistic entering Eq. (3) for such a path is controlled by λ . The parameter ϵ moves the objective only through the incidence of jump paths, which the jump-active conditioning removes, and through the small fraction of multi-episode paths. This weak dependence is consistent with the tuned $\epsilon = 10^{-4}$ lying at the lower edge of the explored range for SPY and for all three cross-asset tickers, and with the Monte Carlo sensitivity of the tuned optima noted in Section S8. The objects entering every expression in this appendix, the fitted boundaries, transition matrix, and residence times, are plotted in Supplementary Fig. S3.

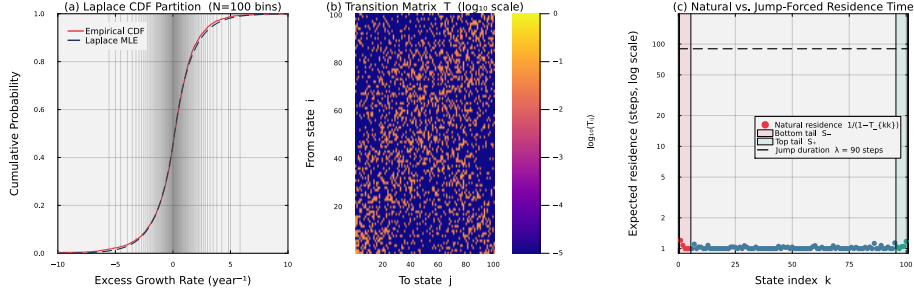


Figure S3: **Fitted hidden Markov model with jumps (HMM-WJ) internals for SPY with $N = 100$ states.** Panel (a) overlays the fitted Laplace cumulative distribution function (CDF) on the empirical CDF; the 99 vertical lines mark the equal-probability quantile boundaries Q_k that define the hidden states. Panel (b) displays the estimated transition matrix $\mathbf{T}$ in $\log_{10}$ color scale; the dominant near-diagonal band reflects short-range regime persistence. Panel (c) plots the expected natural residence time $1/(1 - T_{kk})$ for each state on a $\log_{10}$ scale, with the bottom tail set $\mathcal{S}_-$ (states 1–5, red) and the top tail set $\mathcal{S}_+$ (states 96–100, teal) highlighted. The dashed horizontal line marks the fitted mean jump duration $\lambda = 90$ steps, quantifying the two-order-of-magnitude gap between natural residence times and the jump-duration override.

S5. Transition matrix internals and partition diagnostics

To determine why the ordinary state process required a duration override, we inspected the fitted hidden Markov model with jumps (HMM-WJ) components for SPY at $N = 100$ (Supplementary Fig. S3). The fitted Laplace CDF and the empirical CDF are shown with the 99 state boundaries Q_k . The transition matrix $\mathbf{T}$ has a near-diagonal band, with most off-diagonal mass within ± 5 states of the diagonal. Under the ordinary Markov transitions, every state has an expected residence time of only 1–2 steps. The fitted jump duration is $\lambda = 90$ steps, about two orders of magnitude longer. Every state had at least one in-sample observation. This residence-time gap, rather than missing states, motivated the duration override.

S6. Emission distribution and semi-Markov comparison

To compare the emission and duration specifications, we evaluated three variants on SPY (Supplementary Table S4): a hidden semi-Markov model (HSMM) baseline adapted from Bulla and Bulla (2006) ($K = 8$ states, negative-binomial dwell times with a geometric fallback, and Student- t emissions), HMM-WJ with the original Gaussian emissions ($N = 100$), and HMM-WJ with Student- t emissions ($N = 100$). Both HMM-WJ variants had higher in-sample KS and AD pass rates than the HSMM-style baseline. The HMM-WJ variants use 100 states and the HSMM uses 8, so this comparison does not assign the pass-rate difference solely to the duration model. Holding the rest of HMM-WJ fixed, changing the emission from Normal to Student- t ($\nu = 5$) increased simulated excess kurtosis from 5.5 to 7.5, against an observed value of 7.7 (Supplementary Table S1). The

Table S4: **Three-way comparison of emission distributions and duration specifications for SPY**. Standard errors in parentheses; bold marks the best value per row. The HSMM is a hidden semi-Markov-style model adapted from Bulla and Bulla (2006) ($K = 8$ states, negative-binomial dwell times with a geometric fallback, and Student- t emissions); HMM-WJ: hidden Markov model with jumps ($N = 100$). All entries are computed from 1,000 simulated paths at significance level $\alpha = 0.05$. Kolmogorov–Smirnov (KS), Anderson–Darling (AD), and mean absolute error in the autocorrelation function (ACF-MAE) are defined in Online Appendix S3. Autocorrelation error compares the observed curve with the ensemble-average curve, using lags 1–252 in training and 1–248 in the holdout.

Metric	HSMM (Student- t)	HMM-WJ (Gaussian)	HMM-WJ (Student- t)
States	$K = 8$	$N = 100$	$N = 100$
<i>In-sample (2,766 trading days, 2014–2024)</i>			
KS pass rate (%)	82.0 (1.2)	97.0 (0.5)	98.3 (0.4)
AD pass rate (%)	42.5 (1.6)	90.5 (0.9)	94.8 (0.7)
Excess kurtosis	4.8 (0.08)	5.5 (0.03)	7.5 (0.09)
ACF-MAE	0.059 (<0.001)	0.052 (<0.001)	0.053 (<0.001)
Wasserstein-1	0.176 (0.001)	0.101 (0.002)	0.097 (0.001)
Hellinger	0.113 (<0.001)	0.076 (<0.001)	0.074 (<0.001)
<i>Out-of-sample (249 trading days, 2025)</i>			
KS pass rate (%)	96.2 (0.6)	95.0 (0.7)	95.4 (0.7)
AD pass rate (%)	96.7 (0.6)	95.9 (0.6)	95.3 (0.7)
Excess kurtosis	4.1 (0.13)	4.9 (0.08)	6.2 (0.15)
ACF-MAE	0.042 (<0.001)	0.040 (<0.001)	0.040 (<0.001)
Wasserstein-1	0.287 (0.002)	0.275 (0.005)	0.275 (0.005)
Hellinger	0.239 (0.001)	0.212 (0.001)	0.210 (0.001)

Student- t emission also increased the KS and AD pass rates, while ACF-MAE and the distributional distances changed modestly. The emission choice therefore explained the kurtosis gain in this controlled HMM-WJ comparison, while it changed temporal fit little.

The synthetic grid checked the tracker branch with supplied Gaussian residuals (Table S6). For each target loading and R^2 , the reference return series combined the observed 2,766-day SPY path with an independent Gaussian residual of variance $\sigma_{\varepsilon, \text{target}}^2$ from Eq. (11) and intercept zero. A second Gaussian residual draw was passed directly to Algorithm 2, with $\sigma_{\text{gen}}^2 = \sigma_{\varepsilon, \text{target}}^2$, so the scale in Eq. (12) was one. Only the composed residual was centered over the full path; the reference residual retained its random sample mean. Consequently, the two finite-path constructions were not identical, and realized loading and R^2 estimates could differ from their targets. No full-return hidden Markov model was fitted, and no new market path was simulated. This grid therefore checked branch selection and Gaussian parameter recovery without evaluating full-return reuse or temporal fit. The nominal Kolmogorov–Smirnov pass rates were descriptive because the test’s reference calibration was not established for this shared-market, centered-path comparison.

Table S5: **Hybrid composition method broken out by branch.** N_{tickers} : number of tickers assigned to each branch by the $R^2_{\text{preserve}} = 0.80$ threshold. The clipped variance-preserving branch does not trigger anywhere in the universe ($\rho_i < 0.90$ for all tickers) and is omitted. The two trackers in the R^2 -preserving branch (QQQ and SPYG) had median absolute errors of 0.005 for β and 0.001 for R^2 , with a KS pass rate of 100%.

Branch	N_{tickers}	$ \hat{\beta} - \beta $	$ \hat{R}^2 - R^2_{\text{real}} $	KS pass (%)	W_1	κ
hybrid	421	0.018	0.021	73.5	0.250	7.142
r2-preserve	2	0.005	0.001	100.0	0.099	11.525

Table S6: **Restricted Gaussian check of the R^2 -preserving branch.** Both series used the observed SPY path and independent Gaussian residual draws; the composer used unit residual scale and centered its draw. No full-return model was fitted. Entries are medians over 100 replications per cell with cross-replication standard deviations in parentheses. Loading and R^2 were recovered by ordinary least squares. The nominal Kolmogorov–Smirnov (KS) pass rate is the percentage of comparisons with $p > 0.05$; its reference calibration was not established for this design. Every cell used the R^2 -preserving branch.

β_{true}	R^2_{true}	$\hat{\beta}$	$\hat{R}^2$	Nominal KS pass (%)
0.8	0.80	0.8006 (0.0072)	0.7998 (0.0044)	100.0
0.8	0.85	0.8006 (0.0059)	0.8497 (0.0041)	100.0
0.8	0.90	0.8004 (0.0048)	0.8999 (0.0028)	100.0
0.8	0.95	0.7995 (0.0033)	0.9500 (0.0013)	100.0
0.8	0.99	0.7999 (0.0016)	0.9901 (0.0003)	100.0
1.0	0.80	1.0003 (0.0089)	0.8017 (0.0052)	100.0
1.0	0.85	0.9994 (0.0084)	0.8497 (0.0039)	100.0
1.0	0.90	0.9999 (0.0065)	0.8999 (0.0026)	100.0
1.0	0.95	1.0005 (0.0036)	0.9502 (0.0013)	100.0
1.0	0.99	1.0000 (0.0018)	0.9900 (0.0003)	100.0
1.2	0.80	1.2007 (0.0126)	0.8001 (0.0048)	100.0
1.2	0.85	1.2011 (0.0105)	0.8503 (0.0043)	100.0
1.2	0.90	1.1996 (0.0081)	0.8994 (0.0027)	100.0
1.2	0.95	1.1995 (0.0052)	0.9498 (0.0014)	100.0
1.2	0.99	1.1998 (0.0025)	0.9900 (0.0003)	100.0

S7. State-resolution sensitivity

To test whether the results depended on state resolution, we swept the partition over $N \in \{30, 60, 90, 100, 150, 200\}$ and re-evaluated the in-sample Kolmogorov–Smirnov (KS) and Anderson–Darling (AD) pass rates and the mean absolute error in the absolute-return autocorrelation function (ACF-MAE) for the hidden Markov model with no jumps (HMM-NJ) and the hidden Markov model with jumps (HMM-WJ), holding the tail-set size N_{tail} fixed (Supplementary Table S7). HMM-NJ was insensitive to the partition: its pass rates moved by less than one percentage point and its ACF-MAE was unchanged at the reported precision, as expected for a model with no duration override whose behavior

Table S7: **Sensitivity of in-sample fit to the number of states N .** All entries are computed from 1,000 simulated in-sample-length paths for SPY at significance level $\alpha = 0.05$, with the tail-set size N_{tail} and the remaining hyperparameters of Table 1 held fixed and the (ϵ, λ) grid search re-run at each N for the hidden Markov model with jumps (HMM-WJ). Standard errors in parentheses: binomial for pass rates, bootstrap ($B = 500$) for the mean absolute error of the ensemble-average absolute-return autocorrelation function (ACF-MAE), using lags 1–252. HMM-NJ denotes the hidden Markov model with no jumps. Values at $N = 100$ reproduce the corresponding entries of Table 2.

Model	N	KS pass (%)	AD pass (%)	ACF-MAE
HMM-NJ	30	99.3 (0.3)	97.8 (0.5)	0.059 (<0.001)
HMM-NJ	60	99.5 (0.2)	98.5 (0.4)	0.059 (<0.001)
HMM-NJ	90	99.3 (0.3)	98.6 (0.4)	0.059 (<0.001)
HMM-NJ	100	99.3 (0.3)	98.2 (0.4)	0.059 (<0.001)
HMM-NJ	150	99.1 (0.3)	98.7 (0.4)	0.059 (<0.001)
HMM-NJ	200	99.2 (0.3)	98.3 (0.4)	0.059 (<0.001)
HMM-WJ	30	98.4 (0.4)	97.1 (0.5)	0.057 (<0.001)
HMM-WJ	60	96.7 (0.6)	91.4 (0.9)	0.053 (<0.001)
HMM-WJ	90	98.3 (0.4)	95.2 (0.7)	0.055 (<0.001)
HMM-WJ	100	98.3 (0.4)	94.8 (0.7)	0.053 (<0.001)
HMM-WJ	150	97.7 (0.5)	91.7 (0.9)	0.052 (<0.001)
HMM-WJ	200	96.0 (0.6)	82.8 (1.2)	0.045 (0.001)

could depend on the width of a tail state. HMM-WJ traded the two metrics against each other. Because N_{tail} is fixed, a finer partition makes the tail set a narrower quantile band, so each forced step lands further out, and ACF-MAE improved from 0.057 at $N = 30$ to 0.045 at $N = 200$. The same narrowing left fewer observations behind each tail state’s emission parameters, and the AD pass rate, which weights tail discrepancies heavily, moved the other way, from 97.1% at $N = 30$ to 82.8% at $N = 200$. The intermediate values were not monotone, which is consistent with re-running the (ϵ, λ) grid search on a finite path ensemble at each N . Finer partitions can produce states with too few observations to estimate their emissions locally. The implemented fallback rules keep those emissions defined (Table S17); missing state-level information does not make the fitted parameters undefined. We adopted $N = 100$ for all subsequent experiments as a midpoint that held the KS pass rate above 98% and the AD pass rate within a few percentage points of its best value while capturing part of the ACF-MAE improvement.

S8. Cross-asset generalization

To test whether the single-asset procedure generalized beyond SPY, we repeated the fitting and evaluation on three stocks: NVDA (semiconductors, high volatility), JNJ (health care, low volatility), and JPM (financials, moderate volatility). Per-ticker tune outputs at ($n_{\text{paths}} = 200$, $\text{seed} = 1234$) were reproducible with these settings (Supplementary Table S10). The objective combines ACF and kurtosis errors with fixed weights. Because kurtosis varies across finite

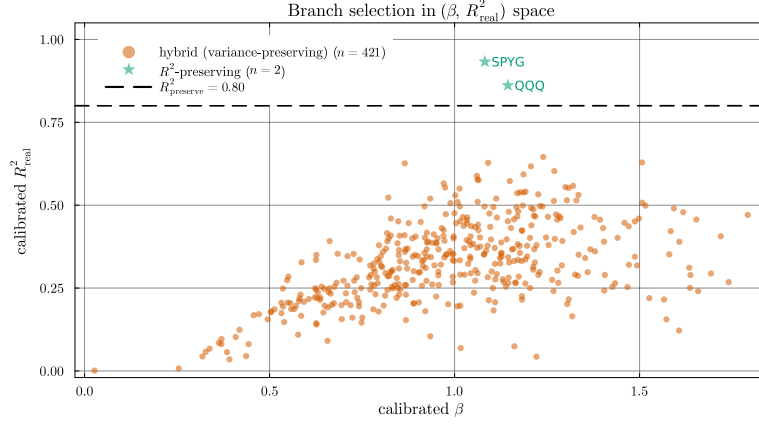


Figure S4: **Branch assignment in calibrated $(\beta, R^2_{\text{real}})$ space.** The $R^2_{\text{preserve}} = 0.80$ threshold (dashed line) separates tracker assets from single stocks. In this universe, only QQQ and SPYG exceed the threshold, and both are assigned to the R^2 -preserving branch. The remaining 421 tickers populate the variance-preserving branch; the clipped variance-preserving branch is unused because the market loading ratio $\rho_i = \beta_i^2 \sigma_m^2 / \sigma_{\text{gen},i}^2$ stays below 0.90 for every ticker.

Table S8: **Aggregate metrics by β quartile, across composition methods.** Quartile cutoffs are computed from the calibrated β distribution across the 423-ticker universe. The hybrid KS pass rate declines from 87.6% in Q1 to 65.9% in Q4 as the market loading ratio ρ_i grows; the naive KS pass rate falls from 24.6% to 1.3% over the same range.

β bucket	Method	$ \hat{\beta} - \beta $	KS pass (%)	W_1	κ	$\text{Var}(g)$
Q1 (low β)	Naive	0.018	24.6	0.403	8.971	18.1
	Gaussian SIM	0.015	0.0	0.620	0.622	14.0
	Hybrid	0.016	87.6	0.177	8.369	14.6
	JumpHMM-on-residuals	0.015	73.2	0.204	7.059	14.3
	Block bootstrap	0.015	71.3	0.208	6.286	13.9
	GARCH(1,1)- t	0.015	57.8	0.253	5.255	13.4
Q2	Naive	0.021	1.9	0.621	7.510	25.1
	Gaussian SIM	0.016	0.2	0.653	1.509	18.0
	Hybrid	0.017	74.9	0.231	6.889	19.0
	JumpHMM-on-residuals	0.017	82.7	0.200	7.799	18.7
	Block bootstrap	0.016	79.8	0.197	7.213	18.0
	GARCH(1,1)- t	0.015	76.4	0.227	6.470	17.8
Q3	Naive	0.023	0.7	0.792	7.390	33.3
	Gaussian SIM	0.017	1.4	0.688	1.947	23.4
	Hybrid	0.018	66.1	0.279	6.904	24.4
	JumpHMM-on-residuals	0.018	78.3	0.242	7.454	24.0
	Block bootstrap	0.017	75.0	0.236	6.888	22.8
	GARCH(1,1)- t	0.016	71.0	0.263	6.265	21.9
Q4 (high β)	Naive	0.029	1.3	0.965	7.233	51.2
	Gaussian SIM	0.022	0.6	0.825	2.054	36.3
	Hybrid	0.022	65.9	0.339	6.739	37.1
	JumpHMM-on-residuals	0.022	69.1	0.313	6.842	37.0
	Block bootstrap	0.021	67.2	0.311	6.334	35.3
	GARCH(1,1)- t	0.021	64.5	0.361	6.064	34.7

Table S9: **Clipping-branch stress test under market-volatility scaling.** We used $g_m^{(\gamma)}(t) = \bar{g}_m + \gamma[g_m(t) - \bar{g}_m]$, with $\gamma \in \{1, 2, 3\}$ and the calibrated marginals held fixed, and re-evaluated the full hybrid composition on the 423-ticker universe. The transformation preserved the market mean and multiplied its standard deviation by γ . The fitted intercepts were retained, so clipping could change asset means according to Eq. (S46).

γ	Clipped (%)
1	0.0
2	76.4
3	95.2

Table S10: **Cross-asset generalization of the hidden Markov model with jumps (HMM-WJ) across three single-stock tickers spanning sector and volatility regimes.** Per-ticker tune outputs (ϵ^*, λ^*) at $n_{\text{paths}} = 200$ and seed = 1234, and the corresponding HMM-WJ Kolmogorov–Smirnov (KS) and Anderson–Darling (AD) pass rates on 1,000 simulated paths at $\alpha = 0.05$ against each ticker’s in-sample (IS; 2014–2024) and out-of-sample (OoS; 2025) growth-rate history. Values in parentheses are binomial standard errors (SEs) for pass rates. The (ϵ^*, λ^*) values are reported as the parameter values used for the IS / OoS metrics rather than as canonical per-ticker optima.

Ticker	ϵ^*	λ^*	IS KS (%)	IS AD (%)	OoS KS (%)	OoS AD (%)
NVDA (semiconductors)	10^{-4}	70	98.9 (0.3)	96.7 (0.6)	97.7 (0.5)	96.1 (0.6)
JNJ (health care)	10^{-4}	70	99.4 (0.2)	96.3 (0.6)	74.0 (1.4)	61.3 (1.5)
JPM (financials)	10^{-4}	80	99.1 (0.3)	95.7 (0.6)	79.2 (1.3)	77.7 (1.3)

samples of heavy-tailed paths, the selected grid point can change with the Monte Carlo sample. The reported (ϵ^*, λ^*) values are reproducible with the stated path count and seed, but they are not unique per-ticker optima. At the selected parameter values, in-sample HMM-WJ KS pass rates were 98.9% for NVDA, 99.4% for JNJ, and 99.1% for JPM. Out-of-sample KS pass rates on the 249-day 2025 hold-out were 97.7%, 74.0%, and 79.2%, respectively. Thus the strong in-sample fit transferred unevenly across assets, especially on the short holdout.

S9. SIM composition derivations and properties verification

To verify the two closed-form targets used by the variance-corrected single-index composition, we derived the variance scale of Eq. (8), the residual-variance target of Eq. (11), and the resulting large- T OLS coefficients and composed variance. To derive Eq. (8), first center the full-return draw as in Eq. (6), then set $\varepsilon_i = s_i \tilde{g}_i^c$ for $s_i \in [0, 1]$ and require that the composed series have the same variance as the generator path:

$$\text{Var}(g_i) = \text{Var}(\alpha_i + \beta_i g_m + \varepsilon_i) = \text{Var}(\beta_i g_m + \varepsilon_i) = \sigma_{\text{gen},i}^2, \quad (\text{S43})$$

where α_i drops out because it is constant. Expanding the sum gives:

$$\text{Var}(\beta_i g_m + \varepsilon_i) = \beta_i^2 \text{Var}(g_m) + 2\beta_i \text{Cov}(g_m, \varepsilon_i) + \text{Var}(\varepsilon_i). \quad (\text{S44})$$

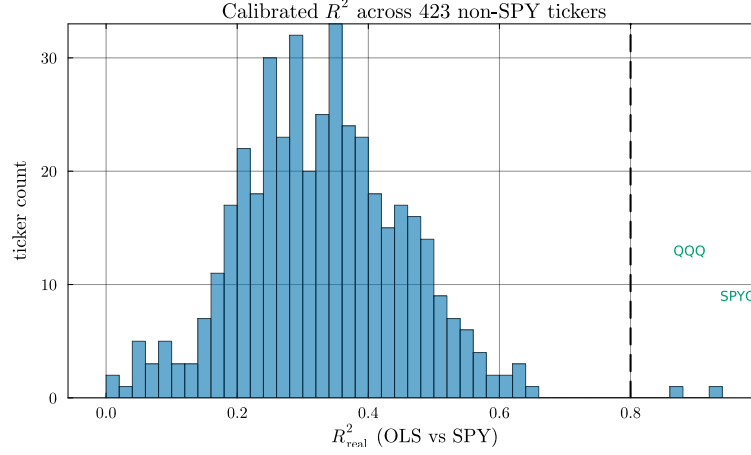


Figure S5: **Calibrated R^2 distribution across the 423 non-SPY tickers.** Histogram of ordinary least-squares (OLS) R^2 against SPY over the 2014–2024 window. The two tickers crossing the $R^2_{\text{preserve}} = 0.80$ threshold (dashed line) are index exchange-traded funds (QQQ, SPYG); the highest- R^2 individual stock (BLK, 0.645) falls ~ 0.16 below the threshold. The R^2 -preserving branch therefore contains only QQQ and SPYG in this universe.

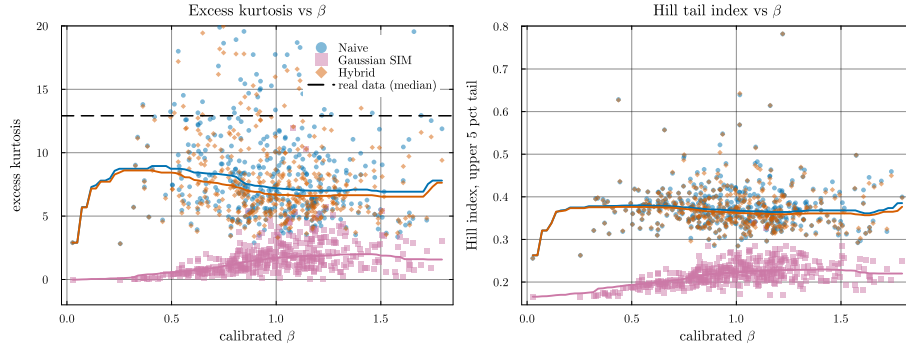


Figure S6: **Heavy-tail preservation: excess kurtosis and Hill tail index vs calibrated β .** Each point is a per-ticker median over 100 replications. *Left:* excess kurtosis, y -axis clipped to $[-2, 20]$ for readability. The dashed line marks the median real-data excess kurtosis across the universe ($\kappa_{\text{real}} \approx 13$); the hybrid and naive composition methods (orange and blue) track the generator's kurtosis, while the Gaussian single-index model (SIM) collapses to $\kappa \approx 1$. *Right:* Hill tail-index estimator on the upper 5% of $|g|$. The hybrid and naive medians remain near 0.37 throughout the β range; Gaussian SIM drops to ~ 0.22 . The hybrid's median being below κ_{real} reflects the JumpHMM marginal's own kurtosis rather than a composition-rule distortion: hybrid tracks naive (both retain the generator's tails) while Gaussian SIM does not.

Substituting $\varepsilon_i = s_i \tilde{g}_i^c$ gives $\text{Var}(\varepsilon_i) = s_i^2 \sigma_{\text{gen},i}^2$ and $\text{Cov}(g_m, \varepsilon_i) = s_i \text{Cov}(g_m, \tilde{g}_i^c)$. Under the assumption $\text{Cov}(\tilde{g}_i^c, g_m) \approx 0$, the cross term vanishes and the variance condition becomes $\beta_i^2 \sigma_m^2 + s_i^2 \sigma_{\text{gen},i}^2 = \sigma_{\text{gen},i}^2$. Solving for s_i^2 gives Eq. (8).

To derive Eq. (11), use the same zero-cross-covariance assumption and write the population R^2 identity for Eq. (4):

$$R^2 = \frac{\beta_i^2 \sigma_m^2}{\beta_i^2 \sigma_m^2 + \sigma_{\varepsilon,i}^2}. \quad (\text{S45})$$

Solving this identity for the residual variance gives Eq. (11). To distinguish intercept retention from mean preservation, let $\bar{g}_m$ be the sample mean of the supplied market path. Centering and scaling give $\bar{\varepsilon}_i = 0$ exactly, and the implementation retains the fitted intercept α_i in every branch. The resulting mean and its change from the calibrated linear mean on the same market path are:

$$\begin{aligned} \bar{g}_i &= \alpha_i + \beta_i^{\text{eff}} \bar{g}_m, \\ \bar{g}_i - (\alpha_i + \beta_i \bar{g}_m) &= (\beta_i^{\text{eff}} - \beta_i) \bar{g}_m. \end{aligned} \quad (\text{S46})$$

Thus, retaining the intercept preserves this linear mean when the loading is unchanged. A clipped path has a different mean whenever $\bar{g}_m \neq 0$. The sign of the shift depends on the loading and market mean; it is not always a downward shift. The variance budget and residual floor determine the clipped loading, with no intercept adjustment to offset this mean change. Retaining α_i also does not imply exact recovery of its OLS estimate from a finite composed path. The finite-sample OLS intercept was:

$$\hat{\alpha}_i = \bar{g}_i - \hat{\beta}_i \bar{g}_m = \alpha_i - (\hat{\beta}_i - \beta_i^{\text{eff}}) \bar{g}_m. \quad (\text{S47})$$

Under the zero-cross-covariance assumption, in the large- T limit OLS therefore gives $\hat{\alpha}_i \rightarrow \alpha_i$ and $\hat{\beta}_i \rightarrow \beta_i^{\text{eff}}$. Here $\beta_i^{\text{eff}} = \beta_i$ unless Eq. (10) clips the market loading. On the variance-preserving branch, the composed variance is $\sigma_{\text{gen},i}^2$. Without clipping, $\text{Var}(g_i) = \beta_i^2 \sigma_m^2 + s_i^2 \sigma_{\text{gen},i}^2 = \sigma_{\text{gen},i}^2$. Under clipping, Eq. (10) sets β_i^{eff} to give the same variance. On the R^2 -preserving branch, the composed variance is $\beta_i^2 \sigma_m^2 / R_{i,\text{real}}^2$, the value implied by the calibrated $(\beta_i, R_{i,\text{real}}^2)$ rather than by the generator marginal. On the ordinary-stock branch, the floor guarantees only that $\text{Var}(\varepsilon_i) \geq f \sigma_{\text{gen},i}^2$; it does not guarantee that adding the market term preserves the generator's tail distribution. The KS pass rate fell from 87.6% in the lowest-beta quartile to 65.9% in the highest-beta quartile (Supplementary Table S8).

The zero-sum constraint also fixes the residual contribution to the terminal cumulative log return. With time step Δt , summing the composed growth rates gives the exact identity:

$$\Delta t \sum_{t=1}^T g_i(t) = T \Delta t \alpha_i + \beta_i^{\text{eff}} \Delta t \sum_{t=1}^T g_m(t), \quad \sum_{t=1}^T \varepsilon_i(t) = 0. \quad (\text{S48})$$

For a fixed market path and effective loading, the residual contributes no terminal uncertainty. Partial sums before T can still vary. If clipping changes the loading between draws, terminal returns can vary through β_i^{eff} , but the residual sum remains zero. Centering uses the full horizon and is not equivalent to subtracting a fitted population mean. Changing the requested horizon can change both the center and the scale applied to an otherwise identical prefix. These paths are therefore constrained over a specified horizon and do not represent unconstrained cumulative residual returns. In the training comparison, the OLS calibration further implies $\alpha_i = \bar{g}_{i,\text{real}} - \beta_i \bar{g}_{m,\text{real}}$, where the bars with subscript real denote observed training means. The difference from the observed asset mean separates into a change in market mean and a change in loading:

$$\bar{g}_i - \bar{g}_{i,\text{real}} = \beta_i(\bar{g}_m - \bar{g}_{m,\text{real}}) + (\beta_i^{\text{eff}} - \beta_i)\bar{g}_m. \quad (\text{S49})$$

Using observed training SPY and $\beta_i^{\text{eff}} = \beta_i$ therefore matches the observed asset mean exactly. The stress transformation preserved the market mean, so only the clipping term could change that mean target. For a supplied market path with a different mean, the original asset mean need not be retained even without clipping. Intercept recovery must be interpreted with these constraints in mind.

S9.1. Role of the centered full-return draw as the residual

The hybrid composition places $\varepsilon_i = s_i \tilde{g}_i^c$, a centered and rescaled draw of asset i 's full return, in the residual slot of Eq. (4). On its face this is a category mismatch: the residual of a single-index decomposition is the idiosyncratic component of the return, orthogonal to the market, while $\tilde{g}_i^c$ is distributed like the centered total return, market-driven component included. This appendix states what the substitution preserves exactly, what it preserves approximately, and how the approximation degrades as the market loading grows.

The residual slot must supply three things: a zero mean, so that the intercept and market term alone set the conditional mean; the variance $\sigma_{\text{gen},i}^2 - \beta_i^2 \sigma_m^2$ not already contributed by the market term; and the asset-specific distributional shape, the heavy tails and volatility clustering that motivated the single-asset generator. Centering gives the residual path an exactly zero sample mean (Eq. (6)). On the unclipped non-tracker branch, using the market path variance in Eq. (8) gives the residual exactly the remaining variance. The composed variance also contains the sample cross-covariance term, which need not vanish on a finite path. Under the independence and large- T assumptions of Section S9, the recovered loadings and intercept converge to their targets. Using the frozen training market variance, as in the six-method holdout evaluation, adds a further variance difference if the simulated market variance differs from that training value. The full-return draw supplies the residual's shape, but its validation as a standalone asset generator does not establish that it reproduces the asset's regression-residual distribution.

At $\rho_i = 0$, Eq. (8) gives $s_i = 1$ and the market term contributes no variance. The composed path then has the same centered empirical distribution as the generator path, while its mean is set by the intercept and market term. Equality

with the original uncentered generator distribution would additionally require those means to agree. For $\rho_i > 0$, the independent market contribution changes even the centered distribution. To describe that change, take the unclipped non-tracker branch with population variances and fixed scales. In population, conditional on the fitted parameters, write the deviation of the composed return from its mean as $X_m + X_i$, where $X_m = \beta_i g_m^c$ is the market contribution, g_m^c is the market path centered at its mean, and $X_i = s_i \tilde{g}_i^c$ is the scaled asset draw; their variances are $v_m = \rho_i \sigma_{\text{gen},i}^2$ and $v_i = (1 - \rho_i) \sigma_{\text{gen},i}^2$. Independence factors every cross term in the fourth moment:

$$\begin{aligned}
\mathbb{E}[(X_m + X_i)^4] &= \mathbb{E}(X_m^4) + 4 \mathbb{E}(X_m^3) \mathbb{E}(X_i) \\
&\quad + 6 \mathbb{E}(X_m^2) \mathbb{E}(X_i^2) + 4 \mathbb{E}(X_m) \mathbb{E}(X_i^3) \\
&\quad + \mathbb{E}(X_i^4) \quad (\text{independence}) \\
&= \mathbb{E}(X_m^4) + 6 v_m v_i + \mathbb{E}(X_i^4). \quad (\mathbb{E}X_m = \mathbb{E}X_i = 0)
\end{aligned} \tag{S50}$$

Writing κ_m and κ_{gen} for the excess kurtosis of the market path and of the generator draw, the component fourth moments are $\mathbb{E}(X_m^4) = (\kappa_m + 3) v_m^2$ and $\mathbb{E}(X_i^4) = (\kappa_{\text{gen}} + 3) v_i^2$, where rescaling by s_i leaves excess kurtosis unchanged. The excess kurtosis of the composed return follows by dividing Eq. (S50) by the squared total variance:

$$\begin{aligned}
\kappa(X_m + X_i) &= \frac{(\kappa_m + 3) v_m^2 + 6 v_m v_i + (\kappa_{\text{gen}} + 3) v_i^2}{(v_m + v_i)^2} - 3 \\
&= \frac{\kappa_m v_m^2 + \kappa_{\text{gen}} v_i^2 + 3 (v_m + v_i)^2}{(v_m + v_i)^2} - 3 \quad (\text{collect terms}) \\
&= \kappa_m \rho_i^2 + \kappa_{\text{gen}} (1 - \rho_i)^2. \quad (v_m + v_i = \sigma_{\text{gen},i}^2)
\end{aligned} \tag{S51}$$

The generator's excess kurtosis survives composition with weight $(1 - \rho_i)^2$ and the market path's enters with weight ρ_i^2 ; a simulation check with independent generator draws matched Eq. (S51) to within 0.03 in mean excess kurtosis at $\rho_i \in \{0.1, 0.3, 0.5\}$ (Supplementary Table S11). For a low-loading asset the composed tails are essentially the generator's: at $\rho_i = 0.10$ the generator's kurtosis retains weight 0.81. As ρ_i grows the composed shape moves away from the asset's own marginal, which is the pattern in the documented diagnostics: the hybrid tracks the generator's kurtosis across the β range while the Gaussian SIM does not (Supplementary Fig. S6), and the hybrid KS pass rate declines from the lowest to the highest β quartile (Supplementary Table S8).

The residual-fit methods offer a direct alternative by fitting each asset's OLS residual series (Table 3). Those fitted series are orthogonal to the observed market by construction, but a new simulated residual path need not have exactly zero sample covariance with its market path. Fitting JumpHMM to residuals requires converting each residual series to a synthetic price path by cumulative compounding and a second fit per asset. The corrected full-return construction

instead reuses the generator’s centered shape and supplies the residual variance budget without that second fit. Its path mean equals $\alpha_i + \beta_i^{\text{eff}} \bar{g}_m$ exactly; matching the observed training mean additionally requires the conditions in Eq. (S49). Variance and factor recovery depend on the stated variance convention and cross-covariance assumptions. At zero market loading it preserves the generator path’s centered empirical distribution. For positive market loading, Eq. (S51) describes how the independent market contribution changes population excess kurtosis on the unclipped non-tracker branch. The clipped and R^2 -preserving branches use the different variance allocations in Algorithm 2.

Table S11: **Simulation check of the kurtosis composition rule.** For each variance share ρ_i , 200 pairs of independent in-sample-length generator paths were standardized and combined as $\sqrt{\rho_i} x + \sqrt{1 - \rho_i} y$. The predicted column evaluates Eq. (S51) with each pair’s sample excess kurtosis; the simulated column is the sample excess kurtosis of the combined series. Both are averaged over the 200 pairs.

ρ_i	Predicted κ	Simulated κ
0.1	6.09	6.11
0.3	4.31	4.34
0.5	3.71	3.74

S10. Uncertainty in the multi-asset jump comparison

The jump comparison used four seeds and 1,000 market replications per window, with all fits frozen (Table 6). Within each setting, naive and corrected composition shared the same asset and market draws. The no-jump and market-only settings reused the no-jump asset draws; the market-only and market-plus-asset settings reused the jump-enabled market draws. Enabled and disabled models used separate random-number streams, so matching replication identifiers did not imply identical innovations. Primary summaries included every path; jump-active subsets only described the mechanism. For each metric, we averaged across tickers within each replication and formed the difference between settings (Table S12). The Monte Carlo standard error was the sample standard deviation of the 1,000 differences divided by $\sqrt{1,000}$, and intervals used the mean difference plus or minus 1.96 standard errors (Table S12). Keeping the cross-section together retained dependence from the shared market draw. The intervals did not treat assets as independent market histories or include uncertainty from fitting or historical-window selection. Kolmogorov–Smirnov (KS) non-rejection remained descriptive because returns were temporally dependent. The temporal contrast between training and holdout remained clear in the replication-level uncertainty estimates (Table S12). The episode frequencies matched the rare-event setting (Supplementary Table S13): 247 of 1,000 training-length SPY paths and 22 of 1,000 holdout-length paths contained an episode. The corresponding shares across enabled asset generators were 24.24% and 2.46%. Market-only jumps reduced average 25-lag autocorrelation error for 99.5% of training assets but only 2.6% of holdout assets. Enabling jumps in both components reduced that

error for 96.9% and 0.5%, respectively. These fractions described ticker-level mean changes, not significance tests. The temporal contrast therefore extended across most assets, within the limits of the short holdout and transferred SPY calibration.

Table S12: **Monte Carlo uncertainty in the effect of jump settings under corrected composition.** Differences are relative to jumps disabled, with 1,000 market replications per window. ΔACF_{25} is the change in mean absolute error of the absolute-return autocorrelation function through lag 25; negative values indicate improved temporal fit. ΔKS is the change in Kolmogorov–Smirnov non-rejection rate in percentage points. Brackets contain approximate 95% Monte Carlo (MC) intervals conditional on the fitted models and observed histories.

Jump setting	ΔACF_{25} [95% MC interval]	ΔKS [95% MC interval]
<i>Training: 2014–2024, 423 assets, 2,766 days</i>		
Market only	−0.00725 [−0.00812, −0.00638]	−0.30 [−1.08, 0.49]
Market + assets	−0.01988 [−0.02063, −0.01913]	−12.25 [−13.02, −11.49]
<i>Holdout: 2025, 416 assets, 249 days</i>		
Market only	0.00195 [0.00104, 0.00287]	−0.04 [−0.41, 0.34]
Market + assets	0.00502 [0.00412, 0.00593]	−1.49 [−1.87, −1.12]

Table S13: **Episode incidence and breadth of the temporal change in the multi-asset jump comparison.** SPY paths with an episode are counted over the 1,000 jump-enabled market replications per window. Asset paths with an episode is the mean jump-active share across the enabled asset generators. The last two columns give the share of assets whose mean absolute-return autocorrelation error through lag 25 fell under corrected composition when jumps were enabled in the market only, or in the market and the assets, relative to jumps disabled.

Window	SPY paths with an episode	Asset paths with an episode (%)	Assets with lower ACF_{25} error, market only (%)	..., market + assets (%)
Training	247 of 1,000	24.24	99.5	96.9
Holdout	22 of 1,000	2.46	2.6	0.5

S11. Residual-centering control

To separate the variance correction from the different residual mean constraints, we paired each native training path with an equivalently centered version (Table S14). The experiment reused the frozen full-return and residual models, ordinary least squares (OLS) calibrations, observed training SPY, seed 1234, and 100 draws per asset of length 2,766. The Gaussian single-index model (SIM), JumpHMM-on-residuals, stationary block bootstrap, and generalized autoregressive conditional heteroskedasticity (GARCH) model retained their native variance scales. With g_i denoting a native composed draw, the control removed only its residual mean:

$$g_i^{\text{ctrl}}(t) = g_i(t) - \frac{1}{T} \sum_{u=1}^T [g_i(u) - \alpha_i - \beta_i^{\text{eff}} g_m(u)] . \quad (\text{S52})$$

The effective loading equaled the fitted loading except when the hybrid rule changed it. Naive and hybrid paths already satisfied the constraint and were reused without modification. No fits, random draws, loadings, or scales changed between a native path and its control. This paired translation left sample variance, recovered loading, R^2 , and excess kurtosis unchanged to numerical tolerance for every evaluated path. The control assessed mean treatment; it did not restore cumulative residual uncertainty or estimate parameter uncertainty.

For a comparison on the same assets, the six-method summaries used the 393 assets whose GARCH fits passed the acceptance checks (Table S14). Refitting all 423 residual series with explicit convergence and admissibility checks reproduced the accepted models' coefficients exactly. All optimizations converged; the same 30 fits were excluded because their ARCH and GARCH coefficients summed to at least one. Acceptance records, coefficient fingerprints, and exclusion reasons accompany the fitted-model cache. The control therefore retained the same models and eligible asset set. There were 39,300 paired paths per method on the common set. The accompanying results also retain each method's full available population: 423 assets for the other five methods and 393 for GARCH. All available-path summaries reproduced the corresponding native values in Table 3 at its reported precision for the metrics included here. Pass rates pooled the same number of draws per asset; Wasserstein distances and intercept errors were medians across paths. Pass rates remained descriptive under serial dependence and the fixed-sum constraint. Equivalent centering favored the residual-fit comparators while preserving the improvement from naive to hybrid composition, so the control narrowed the interpretation of the cross-method comparison without changing the variance correction.

Table S14: **Paired control for residual centering on common training assets.** Native and centered results use the same 393 assets and 100 paths per asset. Centered paths have zero residual mean over 2,766 observations. Kolmogorov–Smirnov (KS) and Anderson–Darling (AD) columns give non-rejection percentages at $\alpha = 0.05$; higher values indicate closer empirical fit. W_1 is median Wasserstein-1 distance, and $|\hat{\alpha} - \alpha|$ is median absolute ordinary least squares (OLS) intercept error, both in yr^{-1} ; lower values indicate closer agreement. Naive and hybrid already center each residual path. JumpHMM denotes the hidden Markov generator, with jumps disabled; SIM denotes the single-index model, and GARCH denotes generalized autoregressive conditional heteroskedasticity.

Method	KS pass (%)		AD pass (%)		W_1		$ \hat{\alpha} - \alpha $	
	Native	Centered	Native	Centered	Native	Centered	Native	Centered
Naive	7.5	7.5	3.7	3.7	0.695	0.695	0.002	0.002
Gaussian SIM	0.6	0.9	0.5	0.6	0.675	0.673	0.047	0.002
Hybrid	74.4	74.4	65.1	65.1	0.246	0.246	0.002	0.002
JumpHMM-on-residuals	75.8	78.4	70.4	71.4	0.230	0.226	0.048	0.002
Block bootstrap	73.5	76.7	66.7	68.0	0.230	0.227	0.041	0.002
GARCH(1,1)- t	67.4	70.0	61.4	63.0	0.267	0.262	0.047	0.002

S12. Pooled one-day tail thresholds

To estimate each generator’s one-day tail without relying on the few extreme observations in a single 249-day path, we pooled returns only after constructing separate paths (Supplementary Table S15). For every frozen training asset and composition method, we generated $B = 5,000$ paths of length 249 and calculated linearly interpolated empirical quantiles from all $249B$ daily log returns. A market path was shared across assets and methods within each replication. Each asset draw retained its own centering and variance correction, with the observed training market variance used in the hybrid rule. The hidden Markov models started from their fitted stationary state distributions, jumps remained disabled, and the generalized autoregressive conditional heteroskedasticity (GARCH) simulator retained its 100-step warmup. We calibrated thresholds for all 423 training assets, including 393 with accepted GARCH fits, before reading the holdout histories. The resulting quantile estimated the return distribution of a uniformly selected day within this finite-horizon construction. It did not estimate a horizon-independent path law or restore cumulative residual uncertainty.

To distinguish numerical stability from holdout performance, we fixed pool sizes of 100, 500, 1,000, 2,500, and 5,000 paths before scoring the observed returns (Supplementary Table S15). An independent ensemble of 1,000 paths per asset and method checked the simulated probability of falling below each threshold. Disjoint random streams separated calibration and validation, while preserving the shared market within each ensemble. Doubling the pool from 2,500 to 5,000 changed the median asset’s 99% threshold by 0.12–0.30%, depending on method. The corresponding 95th percentiles of absolute relative change were 0.37–1.03%. Individual thresholds were less stable than aggregate rates: the largest 99% change was 3.95%, for a GARCH asset. Every method’s mean observed exceedance rate changed by less than 0.006 percentage points at both coverage levels. Mean exceedance rates on independent simulated paths were 5.03–5.04% and 1.01–1.02%, close to the respective 5% and 1% targets. These checks assessed Monte Carlo stability under the fitted generators; they did not establish coverage on real returns. The reported cross-asset standard deviations were not standard errors, and no coverage significance test treated shared market histories or dependent days as independent observations.

Table S15: **Threshold convergence and independent simulation checks.** Relative changes compare pooled thresholds from 2,500 and 5,000 separate 249-day paths, divided by the absolute 5,000-path threshold. Median and P95 denote the cross-asset median and 95th percentile. Simulation rates are mean per-asset exceedance percentages on 1,000 independent paths, using the 5,000-path threshold. Summaries cover the 416 holdout-eligible assets, or 386 for generalized autoregressive conditional heteroskedasticity (GARCH). SIM denotes the single-index model; JumpHMM denotes the hidden Markov generator with jumps disabled.

Method	Relative threshold change (%)		Simulation rate (%)	
	95%: median / P95	99%: median / P95	95%	99%
Naive	0.10 / 0.31	0.28 / 0.90	5.03	1.01
Gaussian SIM	0.08 / 0.24	0.12 / 0.37	5.04	1.02
Hybrid	0.12 / 0.39	0.28 / 0.93	5.03	1.01
JumpHMM-on-residuals	0.11 / 0.31	0.26 / 0.80	5.03	1.01
Block bootstrap	0.14 / 0.45	0.22 / 0.72	5.03	1.01
GARCH(1,1)- t	0.14 / 0.65	0.30 / 1.03	5.03	1.01

To check whether differences in asset eligibility affected the comparison, we restricted all six methods to the same 386 assets with accepted GARCH fits and complete holdout histories (Supplementary Table S16). The corrected method’s 99% exceedance rate was 1.08%, compared with 1.14% for JumpHMM-on-residuals and 0.85% for naive composition. Gaussian single-index composition remained farther above the nominal rate at 1.66%. The common-asset comparison therefore retained the finding that the corrected and residual-fit hidden Markov methods were close to nominal on average. Neither the common-asset nor the full-asset summary established equivalence, precise method rankings, or calibration beyond the single observed market period.

Table S16: **Pooled-threshold coverage on the common holdout assets.** All methods use the same 386 assets. Each threshold pools 5,000 separate 249-day paths, as in Table 5. Rate denotes the mean per-asset exceedance percentage on observed 2025 returns. Standard deviation (SD) describes cross-asset dispersion in percentage points. SIM denotes the single-index model; JumpHMM denotes the hidden Markov generator with jumps disabled; GARCH denotes generalized autoregressive conditional heteroskedasticity.

Method	$\alpha = 0.95$		$\alpha = 0.99$	
	rate (%)	SD (pp)	rate (%)	SD (pp)
Naive	3.95	2.06	0.85	0.69
Gaussian SIM	4.54	2.19	1.66	1.06
Hybrid	5.37	2.43	1.08	0.79
JumpHMM-on-residuals	5.50	2.51	1.14	0.80
Block bootstrap	5.38	2.48	1.25	0.86
GARCH(1,1)- t	5.07	2.45	1.20	0.89

S13. Emission fallback inventory and sensitivity

To identify where the fitted emissions differed from within-state estimates, we reconstructed all state assignments, emission parameters, and counted transition rows from the training inputs (Table S17). For one asset, let n_k be the number of

observations in state k , let $\bar{G}_k$ and s_k be its sample mean and standard deviation when defined, and let $\bar{G}$ and s_G be the corresponding full-series quantities. The implemented emission rule was:

$$(\mu_k, \sigma_k) = \begin{cases} (\bar{G}, s_G), & n_k < 2, \\ (\bar{G}_k, s_G), & n_k \geq 2, s_k < \tau, \\ (\bar{G}_k, s_k), & n_k \geq 2, s_k \geq \tau, \end{cases} \quad \tau = 10^{-12} \text{yr}^{-1}. \quad (\text{S53})$$

The scale σ_k multiplies a standard Student- t draw; it is not the emission standard deviation. State intervals included their lower boundary and excluded their upper boundary. A transition row with no observed outgoing transition was set to $1/N$ in every column. Such a row was unconstrained by the observed transition counts. Reconstruction matched every saved emission parameter and transition probability exactly, including the corresponding fits to regression residuals. Among the 424 full-return models, 39 had at least one fallback emission, affecting 50 states. Of these, 39 states contained 11–119 identical observations each, seven were empty, and four contained one observation. The seven empty states also supplied the seven uniform transition rows. Fallback states carried 0.399–4.300% of stationary probability per affected asset. Sparse-state probability was at most 0.0363% per asset; most fallback mass belonged to populated states with repeated returns. SPY required no fallback, and neither did any of the 423 residual-fit models.

Table S17: **Fallback emissions in the full-return training fits.** Each listed asset had one populated constant-return state containing n_c observations. The columns n_0 and n_1 count empty and singleton states; the other 385 full-return models required no fallback. Fallback mass is $100 \sum_{k \in \mathcal{F}} \bar{\pi}_k$, where $\mathcal{F}$ contains all fallback states for that ticker. For populated constant states, the fitted location remained the state mean while the scale used the asset’s full training standard deviation. Empty and singleton states used both full-series moments. ΔV is the percentage change in stationary mixture variance when only the populated constant-state scales are replaced by their empirical values, with all other parameters fixed. These are changes to the uncomposed generator, not variance errors from composition. All fits used $N = 100$ and $\nu = 5$.

Ticker	n_c	n_0	n_1	Fallback mass (%)	ΔV (%)
ADM	20	0	0	0.723	-1.130
AEE	24	0	0	0.870	-1.361
AES	49	0	1	1.807	-2.792
BMV	19	0	0	0.687	-1.090
CAG	34	0	0	1.231	-1.830
CNP	35	1	0	1.267	-1.906
CSCO	30	0	0	1.087	-1.710
EXC	32	0	0	1.159	-1.728
F	65	1	0	2.349	-3.665
FE	23	0	0	0.828	-1.255
FITB	18	0	0	0.651	-1.025
GLW	28	0	0	1.013	-1.606
HBAN	55	0	1	2.025	-3.120
HRL	34	0	0	1.231	-1.943
HST	43	1	0	1.553	-2.428
IPG	33	0	0	1.194	-1.894
JCI	11	0	0	0.399	-0.651
JNPR	34	0	0	1.230	-1.938
K	15	0	0	0.542	-0.854
KEY	45	0	0	1.627	-2.488
KIM	40	0	1	1.483	-2.219
KMI	36	0	0	1.300	-2.015
KO	28	0	0	1.014	-1.609
MO	30	0	0	1.084	-1.727
NI	36	0	0	1.305	-1.525
NWS	119	1	0	4.300	-6.520
NWSA	59	0	1	2.170	-3.359
O	26	0	0	0.940	-1.391
PPL	44	0	0	1.591	-2.484
RF	49	1	0	1.774	-2.784
ROL	38	0	0	1.371	-2.172
SLV	56	1	0	1.988	-3.148
T	36	0	0	1.302	-2.018
UDR	24	0	0	0.869	-1.374
USB	28	0	0	1.013	-1.602
VZ	28	0	0	1.012	-1.631
WU	56	1	0	2.026	-3.209
WY	28	0	0	1.012	-1.537
XEL	27	0	0	0.976	-1.541

To isolate the consequence of broadening populated constant states, we compared the fitted generators with an empirical-scale alternative on all 39 affected non-market assets (Table S18). The alternative replaced s_G by s_k only

where $n_k \geq 2$ and $s_k < \tau$; locations, partitions, transitions, initial probabilities, and sparse-state fallbacks were unchanged. A zero empirical scale gives a point mass at the state mean. The alternative was a diagnostic of how repeated returns were represented, not a fitted or selected replacement model. We replayed the original 100 paths of 2,766 observations per asset, paired by state sequence and random draws, with jumps disabled and the observed training SPY path shared by both treatments. Returns from unmodified states were identical between treatments. We scored the uncomposed generator and the naive and hybrid compositions; each composition used its treatment’s generator variance. All nine original path metrics for the 7,800 fitted naive and hybrid paths matched the canonical cache exactly. Comparisons with the historical returns used the original scoring functions, so pass fractions remain descriptive summaries rather than calibrated significance claims for dependent series containing ties.

The empirical-scale alternative improved marginal-fit scores on this training subset (Table S18). For hybrid composition, Kolmogorov–Smirnov (KS) pass rates increased from 68.64% to 78.92%, and Anderson–Darling (AD) pass rates increased from 56.15% to 70.23%. The paired changes were 10.28 and 14.08 percentage points, with Monte Carlo standard errors of 0.49 and 0.60 points. Mean Wasserstein distance fell from 0.2782 to 0.2474 yr^{−1}. For the uncomposed generators, the change in stationary mixture variance ranged from −0.651% to −6.520% (Table S17). The fallback scale therefore affected both the generator’s variance and the composed marginal fit. All main comparisons retain the original fallback rule. This paired training check did not test changes to sparse-state estimates or validate the alternative on held-out histories, and it did not establish a preferred production policy.

Table S18: **Paired sensitivity to the scale of populated constant states.** Each treatment used 100 paired training-length paths for each of the same 39 affected assets. Entries are means over these fixed assets and replications. The empirical-scale treatment changes only populated near-constant states, retaining the sparse-state fallback. Kolmogorov–Smirnov (KS) and Anderson–Darling (AD) pass fractions use $p > 0.05$; their changes are percentage points. W_1 is the Wasserstein-1 distance to observed growth rates. Variance / observed divides each path’s variance by its asset’s observed training variance. ACF-MAE₂₅ is the mean absolute difference between each path’s absolute-return autocorrelation function and the observed function over lags 1–25, calculated before averaging paths. Paired change subtracts the fitted value from the empirical-scale value. Standard errors (SE) use the 100 replication-level mean paired changes over the fixed asset set and describe simulation uncertainty, not uncertainty across historical markets.

Output	Metric	As fitted	Empirical scale	Paired change (SE)
Uncomposed	KS pass (%)	98.64	99.82	+1.18 (0.16)
	AD pass (%)	98.90	99.72	+0.82 (0.14)
	W_1 (yr ^{−1})	0.1800	0.1696	−0.0104 (0.0004)
	Variance / observed	1.0772	1.0553	−0.0220 (0.0001)
	ACF-MAE ₂₅	0.14207	0.14183	−0.00024 (0.00002)
Naive	KS pass (%)	12.13	14.64	+2.51 (0.22)
	AD pass (%)	6.92	9.44	+2.51 (0.23)
	W_1 (yr ^{−1})	0.6234	0.5840	−0.0394 (0.0002)
	Variance / observed	1.3650	1.3431	−0.0219 (0.0001)
	ACF-MAE ₂₅	0.11663	0.11576	−0.00086 (0.00002)
Hybrid	KS pass (%)	68.64	78.92	+10.28 (0.49)
	AD pass (%)	56.15	70.23	+14.08 (0.60)
	W_1 (yr ^{−1})	0.2782	0.2474	−0.0308 (0.0002)
	Variance / observed	1.0775	1.0556	−0.0219 (0.0001)
	ACF-MAE ₂₅	0.10289	0.10145	−0.00144 (0.00002)

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